

# Spectral Tree Inference from Incomplete Matrices: Fiedler Vector Bounds under Sub-sampling

Itay Gonen

## Abstract

Top-down reconstruction of latent binary phylogenetic trees is fundamentally bottlenecked by the  $\Theta(m^2)$  cost of computing pairwise similarities across  $m$  leaves. While observing only a sparse, Bernoulli-sampled subset of entries offers a computational workaround, real-world tree imbalance severely degrades the spectral signal, rendering standard balanced-cluster recovery guarantees inadequate. In this work, we frame spectral tree inference as an eigenvector perturbation problem within the matrix-completion regime, demonstrating that the primary root split of an imbalanced latent tree can be exactly recovered from the sign pattern of the empirical Fiedler vector.

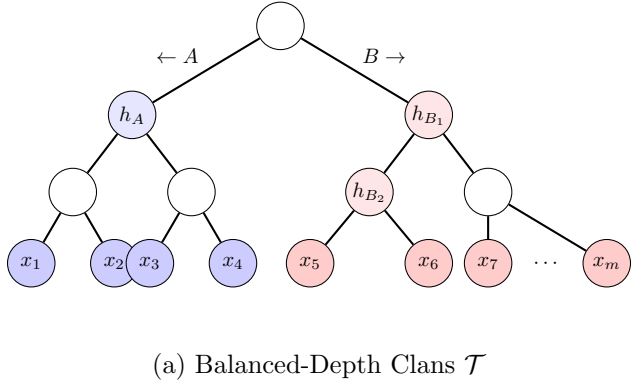
Our primary contribution is a sharp finite-sample guarantee proving that a sampling rate of  $p = \Theta(\log m/m)$  suffices for exact split recovery at a fixed imbalance. By utilizing a coordinate-wise Neumann-series expansion rather than a standard Davis-Kahan  $\ell_2$  transfer, we eliminate a spurious  $\sqrt{m}$  factor in the entry-wise error bound. Furthermore, we identify the imbalance ratio  $\eta$  as the governing topological parameter and prove it dictates recovery through two distinct failure mechanisms: artificially inflating subspace coherence (rendering recovery statistically unaffordable) and eroding the spectral gap (rendering recovery structurally impossible). This yields an explicit sample complexity dependence of  $(1 + \eta)^3$ . Finally, we establish a parallel spectral recovery principle for double-centered distance matrices. Simulations on closed-form block models and Kingman coalescent trees with JC69 sequence evolution validate the predicted  $\Theta(\log m/m)$  scaling and the isolated degradation mechanics of topological imbalance.

## 1 Problem Definition

Our starting point is purely topological. Let  $\mathcal{T} = (V, E)$  be a rooted latent binary phylogenetic tree whose vertex set  $V = \mathcal{X} \cup \mathcal{H}$  splits into  $m$  observed terminal leaves  $\mathcal{X} = \{x_1, \dots, x_m\}$  and  $m - 1$  unobserved internal ancestral nodes  $\mathcal{H} = \{h_1, \dots, h_{m-1}\}$ , among them a distinguished root  $r \in \mathcal{H}$  [2]. The two edges descending from  $r$  define the *primary split* of the tree: the leaf set  $\mathcal{X}$  is partitioned into the two subtrees hanging off the root.

We call each side of this split a *clan* [2, 21, 18]: topologically, a clan is the leaf set of one of the two root subtrees, the maximal monophyletic group induced by the root edge. The term is the unrooted analogue of a clade [21], so the root-split bipartition is precisely a pair of complementary clans, consistent with the unrooted similarity matrix we analyze. This single bipartition  $\mathcal{X} = A \cup B$ , rather than the full branching pattern of  $\mathcal{T}$ , is the object we set out to recover; resolving it is the top-down step from which the remaining tree structure follows recursively. Throughout, we order the leaves along the split so that leaves 1 through  $a$  form clan  $A$  and leaves  $a + 1$  through  $m$  form clan  $B$ , which lets us speak of a clan equivalently as a contiguous block of indices.

The bridge from this topology to linear algebra is the population similarity matrix. Let  $S \in \mathbb{R}^{m \times m}$  be the symmetric matrix of pairwise similarities over the leaves  $\mathcal{X}$ , normalized so that  $S_{ii} = 1$  [2]. Under the clan ordering, the primary split becomes an algebraic object: the within-clan pairs



$$S = \begin{pmatrix} S_A & \beta_0 J_{a \times b} \\ \beta_0 J_{b \times a} & S_B \end{pmatrix}$$

(b) Matrix  $S$  Structure

$$L = \begin{pmatrix} D_A - S_A & -\beta_0 J_{a \times b} \\ -\beta_0 J_{b \times a} & D_B - S_B \end{pmatrix}$$

(c) Laplacian  $L$  Structure

Figure 1: Visual topology of two lineage clans under a localized root split boundary condition, maintaining the same depth but differing terminal node counts. Shown alongside their corresponding population matrix block configurations ( $S$  and  $L$  stacked vertically).

occupy the two diagonal blocks of  $S$ , and the cross-clan pairs its off-diagonal block. Writing  $a = |A|$ ,  $b = |B|$ , and  $J_{a \times b}$  for the all-ones block, the constant cross-clan boundary developed below gives  $S$  and its combinatorial Laplacian  $L = D - S$  the block form shown in Figure 1. The generative model below makes this matrix concrete, turning branch lengths and the substitution process into the block structure of  $S$ .

Branch lengths put a quantitative skin on the bare topology. Each edge  $(h, h') \in E$  carries a branch length  $\tau_{hh'} \geq 0$  representing evolutionary time. Absolute node times  $\tau_h \geq 0$  are measured from the root, with leaves fixed at  $\tau = 0$ , so  $\tau_{hh'} = |\tau_h - \tau_{h'}|$ . Running a substitution process along these edges is what converts the tree into the similarity matrix  $S$ .

**Assumption A1** (Generative model: GTR and log-det similarity). Sequences of length  $\ell$  over a  $d$ -character alphabet evolve along  $\mathcal{T}$  under a *General Time-Reversible* (GTR) rate matrix  $Q \in \mathbb{R}^{d \times d}$  satisfying (i)  $Q_{ij} \geq 0$  for  $i \neq j$ , (ii)  $\sum_j Q_{ij} = 0$  for all  $i$ , (iii) a stationary distribution  $\pi > 0$  with  $\pi Q = 0$ , and (iv)  $\text{diag}(\pi)Q$  symmetric (reversibility). These imply  $\text{tr}(Q) < 0$  with all eigenvalues non-positive, and the substitution matrix along  $(h, h')$  is  $P_{h|h'} = \exp(Q \tau_{hh'})$ . The *population similarity* between leaves  $i, j$  is the log-det similarity [2]

$$S_{ij} = \exp(4 \text{tr}(Q) \tau_{ij}), \quad (1)$$

where  $\tau_{ij}$  is the depth of  $\text{MRCA}(i, j)$ , a purely topological quantity; (1) thus maps the tree onto  $S$ , the first fully linear-algebraic object of the paper. The empirical log-det estimator  $\hat{S}_{ij}$  concentrates around  $S_{ij}$  at rate  $\mathcal{O}(1/\sqrt{\ell})$  [2]; the structural conditions that follow are stated as properties of the population  $S$ , whose observable proxy is  $\hat{S}$ .

We now read the topology off the matrix. Since  $\text{tr}(Q) < 0$ , (1) makes  $S_{ij}$  a decreasing function of the MRCA depth  $\tau_{ij}$ , so leaves with a shallower common ancestor are more similar. Within-clan pairs satisfy  $\tau_{ij} < \tau_{\text{root}}$  and therefore dominate every cross-clan pair: the primary split appears as two strong diagonal blocks of  $S$  separated by a weaker off-diagonal block. Four conditions turn this depth monotonicity into algebra: the molecular clock (Assumption A2) collapses the cross-clan block to a single baseline; the Constant Block Model (Assumption A3) idealizes the two diagonal

blocks as uniformly dominating that baseline; the Hierarchical Block Model (Assumption A4) *relaxes* that idealization to a geometric decay with path depth; and range restriction (Assumption A5) confines every entry to  $(0, 1)$ .

**Assumption A2** (Molecular clock). The tree is ultrametric: all cross-clan MRCA depths equal the root depth, so every cross-clan similarity collapses to a common baseline  $\beta_0 \in (0, 1)$ .

**Assumption A3** (Constant Block Model (CBM)). The population matrix  $S$  satisfies a localized Constant Block Model at the root split if cross-clan similarities equal the baseline  $\beta_0$  while intra-clan similarities strictly dominate it ( $S_{ij} > \beta_0$  for all same-clan  $i, j$ ) [2]. Under a valid CBM configuration,  $S$  and  $L = D - S$  accept the block form of Figure 1, with diagonal sub-degrees  $D_A = \text{diag}(S_A \mathbf{1}_a) + b\beta_0 I_a$  and  $D_B = \text{diag}(S_B \mathbf{1}_b) + a\beta_0 I_b$ . The leading rank-2 invariant subspace  $U = [v^{(1)}, v^{(2)}]$  of  $L$  has  $v^{(1)} = \frac{1}{\sqrt{m}} \mathbf{1}_m$  and Fiedler vector  $v^{(2)}$  piecewise constant on the split.

The CBM collapses all within-clan structure to a single baseline, which holds exactly under the strict molecular clock with *symmetric* clans. For asymmetric, caterpillar-like clans the within-clan MRCA depths vary even under the clock, so intra-clan entries are no longer constant. The next assumption records the corresponding relaxation. It is a *weakening* of Assumption A3, not an extra requirement: the main development proceeds entirely under the CBM, and the HBM is invoked only where the per-edge decay rate  $\alpha$  is traced through the spectral gap (Section 2, Section D).

**Assumption A4** (Hierarchical Block Model with geometric decay). The HBM relaxes the CBM by letting intra-clan similarities decay geometrically with path depth, parameterized by the maximum tree depth  $D_{\max}$ , a per-edge decay factor  $\alpha \in (0, 1]$  (under GTR with unit edges  $\alpha = \exp(4 \text{tr}(Q) \Delta \tau)$ ), and a top-level baseline  $S_{\text{in}}$ . We say  $S$  satisfies the HBM if the weakest intra-clan similarity is bounded below by  $S_{\text{in}}^{\min} \geq S_{\text{in}} \alpha^{D_{\max}}$ . The depth  $D_{\max}$  is governed by tree shape: under Yule or Kingman models it scales as  $\mathcal{O}(\log m)$ , while degenerate caterpillar geometries reach  $\mathcal{O}(m)$  [19, 6].

Table 1 places the two block models in the chain that runs from the generative model to the block structure of  $S$ .

| Generative assumption               | Consequence for $S$                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| GTR alone                           | $S_{ij} = \exp(4 \text{tr}(Q) \tau_{ij})$ ; diagonal (intra-clan) blocks dominate off-diagonal blocks by MRCA-depth monotonicity. |
| GTR + molecular clock               | Localised CBM: cross-clan similarities decay uniformly to an exact baseline $\beta_0$ .                                           |
| GTR + clock + abstracted intra-clan | Full CBM: within-clan structure collapses to a single dominant baseline; applies to symmetric and asymmetric trees.               |
| GTR + clock + path-depth variance   | HBM: intra-clan similarities decay geometrically as $\alpha^{\text{depth}}$ , weakest bounded at depth $D_{\max}$ .               |

Table 1: From the generative model to the block structure of  $S$ .

**Assumption A5** (Bounded / range-restricted entries). All similarities are range restricted,  $0 < S_{ij} < 1$ .

With  $S$  and its block structure in hand, the analysis becomes entirely linear-algebraic, and turns on three scalar parameters. The first is purely topological. With clan sizes  $a = |A| \leq b = |B|$  and

$a + b = m$ , the *imbalance ratio* measures how unevenly the root split divides the leaves,

$$\eta := \frac{b}{a} = \frac{|B|}{|A|} \geq 1, \quad (2)$$

so that  $\eta = 1$  is the perfectly balanced split while caterpillar-like lineages reach  $\eta = \mathcal{O}(m)$ ; this parameter plays the role of the balanced-cluster constant of Balakrishnan et al. [4]. The second measures how sharply intra-clan similarities exceed cross-clan ones. Writing  $S_{\text{out}}^{\max}$  and  $S_{\text{out}}^{\min}$  for the maximum and minimum similarities across the primary root split, and following the range-restriction framework of [4], we capture the sharpness of the structural partition by the *structural margin*

$$\rho := (S_{\text{in}}^{\min} - S_{\text{out}}^{\max}) - (S_{\text{out}}^{\max} - S_{\text{out}}^{\min}). \quad (3)$$

Under the molecular clock (Assumption A2) cross-clan variance vanishes,  $S_{\text{out}}^{\max} = S_{\text{out}}^{\min} = \beta_0$ , the penalty term collapses, and the parameter reduces to  $\rho = S_{\text{in}}^{\min} - \beta_0$ . The third measures how concentrated the leading invariant subspace of  $L$  is across nodes. Letting  $U = [v^{(1)}, v^{(2)}] \in \mathbb{R}^{m \times 2}$  span the rank-2 invariant subspace of the two leading eigenvalues of  $L$ , with  $v^{(1)} = \frac{1}{\sqrt{m}} \mathbf{1}_m$  the trivial flat eigenvector and  $v^{(2)}$  the structural Fiedler vector, the *subspace coherence* is

$$\mu(U) := \frac{m}{2} \max_{1 \leq i \leq m} \|P_U e_i\|_2^2 = \frac{m}{2} \max_{1 \leq i \leq m} ((v_i^{(1)})^2 + (v_i^{(2)})^2), \quad P_U = UU^\top. \quad (4)$$

The last two are spectral quantities of  $S$  and  $L = D - S$ ; Section 2 shows that the first determines both.

The three parameters are not free of one another. The margin and the imbalance trade off directly, and the split is recoverable at all only when the former dominates the latter; we record this as a standing assumption, derived from the spectral-gap bound of Lemma 1 in Section 2.

**Assumption A6** (Separation margin / recoverability). Under the CBM, the primary split is spectrally recoverable only if the structural margin strictly dominates the imbalance-weighted cross-clan boundary,

$$\rho > \eta S_{\text{out}}^{\max}, \quad \text{equivalently} \quad S_{\text{in}}^{\min} > (1 + \eta) S_{\text{out}}^{\max}. \quad (5)$$

This keeps the spectral gap positive ( $\Delta\lambda > 0$ ). Under the relaxation of Assumption A4 the geometric floor on  $S_{\text{in}}^{\min}$  plays the same role, with  $\alpha^{D_{\max}} > (1 + \eta) S_{\text{out}}^{\max} / S_{\text{in}}$  in place of (5) (Proposition 9).

In practice the population matrix  $S$  is hidden, and only a sparse sample of its entries is observed. Following the matrix-completion paradigm [9, 8], we observe entries through a random mask and reweight, as recorded in the sampling assumption.

**Assumption A7** (Independent sampling with inverse-probability weighting). Let  $\Omega \in \{0, 1\}^{m \times m}$  be a symmetric mask whose upper-triangular entries are independent Bernoulli( $p$ ),  $p \in (0, 1]$ . The empirical similarity is formed by inverse-probability weighting [7],

$$\hat{S}_{ij} = \begin{cases} \frac{1}{p} \Omega_{ij} S_{ij}, & i \neq j, \\ 1, & i = j, \end{cases} \quad (6)$$

which is unbiased,  $\mathbb{E}[\hat{S}] = S$ , so  $E_L = \hat{L} - L$  is mean zero, where  $\hat{L} = \hat{D} - \hat{S}$ .

The Laplacian is not the only operator on these leaves whose spectrum carries the split. A second construction replaces  $S$  by the pairwise *distance* matrix  $\mathcal{D}$  — distinct from the degree matrix  $D$  of  $L = D - S$  — built from  $S$  by  $\mathcal{D}_{ij} = -\log S_{ij}$  (Section F.1), and centers it,  $\mathcal{B} := H\mathcal{D}H$

with  $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ . There the split sits in the sign pattern of  $u^{(1)}$ , the eigenvector of  $\mathcal{B}$  at the eigenvalue of largest absolute value, rather than in the Fiedler vector of  $L$ . Section F works out the population spectrum of  $\mathcal{B}$  and shows that the same Bernoulli( $p$ ) mask of Assumption A7 applies to  $\mathcal{D}$  unchanged, so both operators are observed under one sampling model and Section 3 states a rate for each.

**Problem.** Given the sparsely observed estimate  $\hat{S}$ , recover the primary split  $(A, B)$  from the sign pattern of the Fiedler vector  $\hat{v}^{(2)}$  of  $\hat{L}$ . We ask for the minimal sampling rate  $p$  at which this top-down partition is correct with high probability. Since  $\mathbb{E}[\hat{S}] = S$ , the question reduces to an eigenvector-perturbation problem over  $E_L = \hat{L} - L$ , which the remainder of the paper analyzes.

Recovery is, at its core, an *entry-wise* event. The top-down partition succeeds precisely when the empirical Fiedler vector agrees in sign with its population counterpart at every node, that is, when

$$\left\| \hat{v}^{(2)} - v^{(2)} \right\|_\infty < \min_i |v_i^{(2)}|. \quad (7)$$

Condition (7) is the formal recovery event that every bound in the paper is measured against.

## 2 Topological Scaling and Spectral Gap

The reader now has the population model and the recovery problem. What remains is to explain *why* the topology of the primary split governs recovery. We answer this through the single structural parameter  $\eta$  of (2) and trace its effect on the two spectral quantities the main theorem depends on. Working with the clean population Laplacian  $L = D - S$ , both the subspace coherence  $\mu(U)$  and the spectral gap  $\Delta\lambda = \lambda_3 - \lambda_2$  are fixed by macro-topological features alone, and  $\eta$  moves both against recovery: it first inflates  $\mu(U)$ , then erodes  $\Delta\lambda$ .

**Proposition 1** (Topological scaling of coherence). *Under Assumption A3, the tree imbalance  $\eta$  maps deterministically to the subspace coherence,*

$$\mu(U) = \frac{1 + \eta}{2}. \quad (8)$$

The identity follows by substituting the Fiedler coordinates of Lemma 4 into (4); the short computation is given in Section A. Thus a balanced split attains the optimal coherence  $\mu(U) = 1$ , and the signal spreads more unevenly across nodes as  $\eta$  grows; the generated regime of Section 5 confirms the identity on data from the full generative pipeline.

The second spectral quantity is the gap  $\Delta\lambda = \lambda_3 - \lambda_2$  that isolates the Fiedler vector, and the same imbalance that raises  $\mu(U)$  drives it toward zero.

**Lemma 1** (Spectral-gap lower bound). *Let  $L$  be the combinatorial Laplacian of a similarity matrix satisfying the molecular clock, with structural margin  $\rho = S_{\text{in}}^{\min} - S_{\text{out}}^{\max}$ . For  $m_{\min} = \min(a, b)$  and imbalance  $\eta$ , the spectral gap isolating the Fiedler vector satisfies*

$$\Delta\lambda \geq m_{\min}\rho - \eta m_{\min} S_{\text{out}}^{\max}. \quad (9)$$

The full algebraic-connectivity derivation is given in Section A, building on the eigenpairs of Lemma 4. The bound (9) is purely a CBM statement, controlling the gap through the abstract minimum intra-clan similarity  $S_{\text{in}}^{\min}$ . Under the relaxation of Assumption A4 that minimum stops being a free parameter and becomes a function of tree shape: it is pinned to the geometric floor

$S_{\text{in}} \alpha^{D_{\text{max}}}$ , so depth erodes the gap in the same way imbalance does, and the gap survives only while the decay  $\alpha^{D_{\text{max}}}$  still clears the imbalance-weighted boundary — quantified in Proposition 9.

Proposition 1 and Lemma 1 show that the same parameter  $\eta$  acts on both spectral quantities at once, which produces two distinct ways for top-down recovery to fail; together they bound the admissible region of  $(m, \eta)$ . The first is *structural collapse*,  $\Delta\lambda \rightarrow 0$ , which we state here as a corollary of the gap lemma; the second is *statistical infeasibility*,  $p > 1$ , which is a consequence of Theorem 1 below. A positive gap requires the structural margin to dominate the imbalance-weighted cross-clan boundary,  $m_{\text{min}}\rho > \eta m_{\text{min}}S_{\text{out}}^{\text{max}}$ , i.e.  $\eta < S_{\text{in}}^{\text{min}}/S_{\text{out}}^{\text{max}}$  — precisely what Assumption A6 records. Translating this ratio into tree distances bounds the imbalance the topology can sustain.

**Corollary 1** (Structural imbalance tolerance). *Write similarities as distances,  $S_{ij} = e^{-d(x_i, x_j)}$ , and let  $r(\mathcal{T}) = \max_{i,j}(-\log S_{ij})$  be the diameter and  $h(\mathcal{T})$  the depth of  $\mathcal{T}$  [2], with  $h(\mathcal{T}) < r(\mathcal{T})$ . Then a positive spectral gap is possible only if*

$$\eta \leq \mathcal{O}(e^{\mathcal{O}(h(\mathcal{T})) - r(\mathcal{T})}). \quad (10)$$

*In particular, for an average Kingman coalescent tree, where  $r(\mathcal{T}), h(\mathcal{T}) = \mathcal{O}(\log m)$  [12, 3], this gives  $\eta \leq \mathcal{O}(m^c)$  for a model-dependent constant  $c$ .*

The proof sketch (substituting the diameter/depth bounds into  $\eta < S_{\text{in}}^{\text{min}}/S_{\text{out}}^{\text{max}}$ ) is given in Section A.

So under balanced generative models the gap survives polynomial imbalance. A positive gap is necessary but not sufficient, however: even when  $\Delta\lambda > 0$ , finite sampling may demand more entries than exist, which is the second failure mode. Inflated coherence  $\mu(U) = (1 + \eta)/2$  concentrates the split signal on a vanishing fraction of nodes, so uniform sampling must observe an ever-larger share of  $S$  to capture it. Theorem 1 makes this precise: the rate its sufficient condition demands grows with  $\eta$ , and once that demand exceeds  $p = 1$  the guarantee is vacuous — Corollary 3 locates the threshold. This barrier is the coherence-limited sampling phenomenon familiar from matrix completion, where recovery becomes infeasible once the leading subspace concentrates on too few coordinates [9, 17, 10], mirroring the noise threshold for spectral clustering [4]. The two bounds frame the admissible region:  $\eta$  must be small enough for the gap to open (structural) and small enough for sampling to remain feasible (statistical).

### 3 Main Result

The entry-wise recovery criterion (7) is what the sampling rate has to buy. This section converts it into a sufficient condition on  $p$ , first for the Laplacian  $L = D - S$  and then for the centred distance operator  $\mathcal{B}$ , stated throughout in the structural quantities of Section 2: the margin  $\rho$  of (3) and the imbalance  $\eta$  of (2).

**Theorem 1** (Exact top-down split recovery). *In a non-balanced CBM with  $m$  terminal nodes and imbalance  $\eta \geq 1$ , let  $\rho$  be the structural margin and let  $\hat{v}^{(2)}$  be the empirical Fiedler vector — the unit eigenvector belonging to the second-smallest eigenvalue of the sampled Laplacian  $\hat{L} = \hat{D} - \hat{S}$ , formed from entries observed with probability  $p$  as in Assumption A7. Assume Assumptions A1 to A3 and A5 to A7. For the top-down partition to succeed (i.e.  $\|\hat{v}^{(2)} - v^{(2)}\|_{\infty} < \min_i |v_i^{(2)}|$ ), it suffices that*

$$p > \frac{8(1 + \eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\text{max}})^2} = \Theta\left(\frac{\log m}{m}\right), \quad (11)$$

*in which case  $\text{sign}(\hat{v}^{(2)})$  recovers  $(A, B)$  exactly with high probability.*

The distance operator  $\mathcal{B} = HDH$  of Section 1 admits a statement of the same shape — entry-wise sign agreement of an empirical structural eigenvector against a piecewise-constant target, at a rate governed by a spectral gap. Since the hypotheses of the two operators are not nested, the following is a *sibling* of Theorem 1 rather than a corollary of it.

Two of the population facts that the Laplacian route *has* — a piecewise-constant structural vector (Lemma 4) and a gap bound (Lemma 1) — are unavailable for  $\mathcal{B}$  at general imbalance (Remark 7), so they are hypotheses below rather than consequences. A third hypothesis, control of the second-order term of the Neumann expansion, is stated explicitly here; it is needed on both routes and is discharged on neither (Remark 1). What is *not* needed is any assumption about within-clan cancellation: the per-node variance is bounded by  $d_0^2/p$  using only  $\|u^{(1)}\|_2 = 1$  and  $\mathcal{D}_{ij} \leq d_0$  (Lemma 13).

**Theorem 2** (Distance-route sampling rate). *Let  $\mathcal{D} \in \mathbb{R}^{m \times m}$  be the tree-additive pairwise distance matrix built from  $S$  by  $\mathcal{D}_{ij} = -\log S_{ij}$  (Section F.1), let  $\mathcal{B} = HDH$ , and let  $u^{(1)}$  be a unit-norm eigenvector of  $\mathcal{B}$  at the eigenvalue of largest absolute value. Let  $\hat{\mathcal{D}}$  be the inverse-probability weighted sub-sample of  $\mathcal{D}$  obtained by applying Assumption A7 to  $\mathcal{D}$  (Section F.6), and let  $\hat{u}^{(1)}$  be a unit-norm eigenvector of  $\hat{\mathcal{B}} = H\hat{\mathcal{D}}H$  at its eigenvalue of largest absolute value, signed so that  $\langle \hat{u}^{(1)}, u^{(1)} \rangle \geq 0$ . Assume Assumptions A1 to A3, A5 and A7, and suppose*

- (i) *the top spectral gap  $\Delta\lambda^{(\mathcal{B})} = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$  is strictly positive;*
- (ii)  *$u^{(1)}$  is piecewise constant on the split  $(A, B)$ , with the two values it then takes given by Lemma 13;*
- (iii) (remainder control)  *$\|E_{\mathcal{B}}\|_2 \leq \varepsilon_m \Delta\lambda^{(\mathcal{B})}$  with  $\varepsilon_m \leq c_0(m\eta)^{-1/4}$  for an absolute constant  $c_0$ .*

*There are universal constants  $C, c > 0$  such that the sign-recovery condition (26) holds for  $\hat{u}^{(1)}$  — and hence  $\{\hat{A}, \hat{B}\} = \{A, B\}$  whenever the population sign rule (25) returns  $(A, B)$  — with probability at least  $1 - O(m^{-c})$  whenever*

$$p > \frac{C d_0^2 \eta m \log m}{(\Delta\lambda^{(\mathcal{B})})^2}, \quad d_0 = -\log \beta_0. \quad (12)$$

*Remark 1* (What remainder control costs, on either route). Hypothesis (iii) is the step at which both routes are currently incomplete, and it is stated rather than derived for that reason. The recovery event (26) compares an  $\ell_\infty$  error against a margin of order  $(m\eta)^{-1/2}$ , while the only available bound on the second-order Neumann term is through the operator norm,  $\|w^{(2)}\|_\infty = O(\|E_{\mathcal{B}}\|_2^2 / (\Delta\lambda^{(\mathcal{B})})^2)$  (Lemma 9), which is  $O(\varepsilon_m^2)$  and therefore  $o((m\eta)^{-1/2})$  precisely under (iii). Combining (iii) with the operator-norm bound of Lemma 12 requires  $p \gtrsim d_0^2 m^{3/2} \eta^{1/2} \log m / (\Delta\lambda^{(\mathcal{B})})^2$ , which in the balanced flat CBM is  $\Theta(\log m / \sqrt{m})$  — a factor  $\sqrt{m}$  above (13). Removing the gap between the two requires an entry-wise, rather than  $\ell_2$ -derived, bound on  $w^{(2)}$ , of the leave-one-out type used by [1]. The same gap is present in Theorem 1: at  $p = \Theta(\log m / m)$  the ratio  $\|E_L\|_2 / \Delta\lambda$  implied by Lemmas 1 and 5 is bounded but does not vanish with  $m$ , so the remainder is not  $o(\min_i |v_i^{(2)}|)$  there either. Neither theorem’s stated rate is therefore self-contained without (iii) or its Laplacian analogue.

Both theorems are proved by the same three stages and synthesis — population geometry, a Neumann reduction, per-node concentration, and the comparison against the entry-wise criterion — assembled in Section 4, the distance case substituting  $L \rightarrow \mathcal{B}$ ,  $v^{(2)} \rightarrow u^{(1)}$ ,  $E_L \rightarrow E_{\mathcal{B}}$  and running in full in Sections F.6 and F.7. The two statements differ in how far their parameters can be resolved. Equation (11) is closed-form in the model, because the gap bound  $\Delta\lambda \geq m_{\min}(\rho - \eta S_{\text{out}}^{\max})$  of Lemma 1 can be substituted. No lower bound on  $\Delta\lambda^{(\mathcal{B})}$  valid at general imbalance is available (Remark 7), so (12) is left in terms of the abstract gap. Where the gap is known, the two rates coincide in shape.

**Corollary 2** (Distance rate in the balanced flat CBM). *Under the hypotheses of Theorem 2 in the balanced flat CBM of Proposition 10 —  $|A| = |B| = m/2$ , within-clan distance  $d_1$ , cross-clan distance  $d_0 > d_1$ , a configuration realized by two stars joined by an edge rather than by a binary tree (Remark 6) — hypotheses (i) and (ii) hold with  $\Delta\lambda^{(B)} = m(d_0 - d_1)/2$  and  $\eta = 1$ , so (12) reads*

$$p > \frac{4C d_0^2 \log m}{m(d_0 - d_1)^2} = \Theta\left(\frac{\log m}{m}\right), \quad (13)$$

*the same exponent as (11), with the similarity-side ratio  $\beta_0^2/(\rho - \eta S_{\text{out}}^{\max})^2$  replaced by  $d_0^2/(d_0 - d_1)^2$ , and subject to hypothesis (iii) (Remark 1). This is the only configuration in which hypotheses (i) and (ii) are verified; at general imbalance, and on binary trees, they remain open.*

What remains is to record where these sufficient conditions can be met at all.

**Corollary 3** (Vacuity of the certificate under imbalance). *Under the hypotheses of Theorem 1, with  $\beta_0$  and the margin  $\rho - \eta S_{\text{out}}^{\max}$  held bounded away from 0 independently of  $\eta$ , the right-hand side of (11) exceeds 1 once  $\eta = \Omega(\sqrt[3]{m/\log m})$ ; in general the threshold is reached when  $\eta^3 \beta_0^2/(\rho - \eta S_{\text{out}}^{\max})^2 = \Omega(m/\log m)$ . No admissible sampling rate  $p \in (0, 1]$  then satisfies the sufficient condition, so Theorem 1 certifies nothing about the split in that regime.*

*Remark 2* (Coherence-limited sampling). Corollary 3 is the statistical failure mode previewed in Section 2: holding the structural margin away from collapse ( $\rho - \eta S_{\text{out}}^{\max} > 0$ ) and isolating  $\eta$  in (11) yields the stated threshold. The mechanism is localisation of the signal. As  $\eta$  grows the split concentrates onto a vanishing subset of nodes, and the inflated coherence  $\mu(U) = (1 + \eta)/2$  (Proposition 1) prevents uniform sampling from capturing the critical entries without observing nearly all of  $S$  — the coherence-limited barrier familiar from matrix completion, where recovery becomes infeasible once the leading subspace concentrates on too few coordinates [9, 17, 10]. This is a statement about the reach of a sufficient condition and not about recovery itself: *we prove no converse*, and nothing here rules out that the Fiedler sign rule, or some other estimator, still recovers  $(A, B)$  beyond the threshold. Read together with the structural bound of Corollary 1, (11) delimits the region of  $(m, \eta)$  in which top-down spectral recovery is *certified* to succeed, not the region in which it is possible. (Section F.4 shows that under Assumption A2 the diameter entering Corollary 1 coincides with the cross-clan distance, which weakens that corollary considerably; the two statements should be reconciled.)

*Remark 3* (Scope of the distance rate, and the price it pays). Hypotheses (i) and (ii) are verified only in the balanced flat CBM (Proposition 10 and remark 7), and hypothesis (iii) nowhere (Remark 1). Where they do hold, the two routes share the exponent  $\Theta(\log m/m)$  and differ in one parameter group: the similarity-side ratio  $\beta_0^2/(\rho - \eta S_{\text{out}}^{\max})^2$  becomes  $d_0^2/(d_0 - d_1)^2$ , larger by a factor 275 at  $\eta = 1$  and the synthesized-regime values  $\beta_0 = 0.05$ ,  $S_{\text{in}} = 0.9$  of Section 5. The two rates' unspecified constants are not compared. This comparison fixes the target split and asks which operator tolerates more noise; which split each rule targets is the separate question of Remark 8, and Section F.8 carries the details of both.

## 4 Proof Outline

The proof of Theorem 1 compares two quantities evaluated at the same node: the entry-wise error  $|\hat{v}_i^{(2)} - v_i^{(2)}|$  and the entry-wise margin  $|v_i^{(2)}|$ . It runs in three stages — population geometry fixes the margin and noise budget, a resolvent expansion reduces the error at each node to a scalar, and



a per-node Bernstein bound controls that scalar — after which imposing (7) yields the rate (11). Each stage is stated as a lemma; the proofs are in Sections A to C.

Two population quantities of  $L$  fix the two sides of (7). The spectral gap  $\Delta\lambda$  is bounded below in Lemma 1 by  $m_{\min}(\rho - \eta S_{\text{out}}^{\max})$ , kept positive by Assumption A6; the margin is read off the piecewise-constant Fiedler vector of Lemma 4 as  $\min_i |v_i^{(2)}| = c_B = 1/\sqrt{m\eta}$ , attained on the larger clan  $B$ . Under imbalance these two extremes sit at different nodes, which is what makes the final synthesis  $\eta$ -dependent. A third quantity is a standing hypothesis: the Neumann expansion introduced next converges, with negligible remainder against  $c_B$ , only in the strong-signal regime  $\|E_L\|_2 < \Delta\lambda$ ; Lemma 5 (Section B) supplies the noise side of it,  $\|E_L\|_2 \leq \tilde{C}\sqrt{m(1+\eta)\log m/p}$ , to be compared against the gap bound above.

Classical eigenvector perturbation bounds, foremost Davis–Kahan, control the error only in  $\ell_2$ ; passing to the  $\ell_\infty$  norm of (7) via  $\|\cdot\|_\infty \leq \|\cdot\|_2$  forfeits a factor  $\sqrt{m}$ , enough under imbalance to push the required rate past  $p \leq 1$  (the rate  $p_{\text{naive}}$  of Remark 5). A coordinate-wise argument is needed instead: following Eldridge et al. [11], Fan et al. [13], and Abbe et al. [1], we expand the perturbed Fiedler vector as a resolvent (Neumann) series in the noise operator  $E_L$ , isolating a single leading term  $-(E_L v^{(2)})_i / \Delta\lambda$  per coordinate with a second-order remainder. That term is a sum of  $O(m)$  independent, mean-zero increments rather than an extremal singular value, so a per-node Bernstein bound concentrates it a factor  $\sqrt{m}$  tighter than a global operator-norm bound would pay — the saving that turns (7) into the tight rate (11) (Section E). It is not a result of this paper; we record it as a lemma.

**Lemma 2** (Neumann decomposition of the Fiedler vector). *Let  $L$  be symmetric with eigenpairs  $\{(\lambda_\ell, v^{(\ell)})\}$ , let  $\hat{L} = L + E_L$  be a symmetric perturbation, and write  $\hat{v}^{(2)} = v^{(2)} + w$ . If  $\|E_L\|_2 < \Delta\lambda$ , then, with  $\Pi$  the spectral projector onto the complement of  $v^{(2)}$ ,  $w$  admits the exact resolvent expansion  $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$ , whose first-order term is*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad (14)$$

so that entrywise  $w_i = -(E_L v^{(2)})_i / \Delta\lambda + R_i$  with the remainder of second order,  $\|R\|_\infty = O(\|E_L\|_2^2 / \Delta\lambda^2)$ .

The decomposition and its first-order bound are proved in Section C.1 (Lemmas 6 and 7, Remark 4). The second-order remainder  $R$  is bounded separately in Section C.3 (Lemma 9); its discard is what confines the argument to the strong-signal regime identified above.

Lemma 2 leaves a scalar,  $(E_L v^{(2)})_i$ , and Section C.2 shows it is well-behaved: Lemma 8 rewrites it as a sum of independent, mean-zero increments  $X_{ij} = (\omega_{ij}/p - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$ , Proposition 5 removes every within-clan pair since  $v^{(2)}$  is constant on each clan, and Proposition 6 evaluates the surviving variance on clan  $B$  at  $\sigma_B^2 = (1-p)(1+\eta)\beta_0^2/(\eta p) = (1-p)\tilde{\rho}(\eta)/p$  with  $\tilde{\rho}(\eta) = (1+\eta)\beta_0^2/\eta$ . Bernstein’s inequality then applies directly, and since this is the only place  $p$  enters the final condition, we state it in the main text.

**Lemma 3** (Per-node concentration). *With scale  $R_B = \beta_0(1+\eta)/(p\sqrt{m\eta})$ , for each  $i \in B$  and  $\delta > 0$ , with probability  $\geq 1 - \delta$ ,  $|(E_L v^{(2)})_i| \leq 2\sqrt{2\sigma_B^2 \ln(2/\delta)} + \frac{2}{3}R_B \ln(2/\delta)$ . In the variance-dominated regime ( $p = \Theta(\log m/m)$ ), a union bound over all  $m$  nodes at  $\delta = \delta_0/m$  gives, with high probability uniformly over  $i$ ,*

$$|(E_L v^{(2)})_i| \lesssim \sqrt{\frac{8(1+\eta)\beta_0^2 \log(2m/\delta_0)}{\eta p}}. \quad (15)$$

The proof is in Section C.2. This is the step that realises the  $\sqrt{m}$  saving quantified in Remark 5: the variance  $\sigma_B^2$  of the individual increment sum is smaller than  $\|E_L\|_2^2$  by a factor  $m$ . The three stages now combine.

*Proof of Theorem 1. Step 1 (population geometry).* Lemma 1 lower-bounds the spectral gap by  $\Delta\lambda \geq m_{\min}(\rho - \eta S_{\text{out}}^{\max})$ , positive under Assumption A6; Lemma 4 gives the right-hand side of (7) as  $\min_i |v_i^{(2)}| = c_B$  with  $c_B^2 = 1/(m\eta)$ .

*Step 2 (entry-wise reduction).* Lemma 2, applicable in the strong-signal regime of Step 1, writes  $\hat{v}^{(2)} - v^{(2)} = w$  with  $w_i = -(E_L v^{(2)})_i / \Delta\lambda + R_i$  and  $\|R\|_\infty = O(\|E_L\|_2^2 / \Delta\lambda^2)$ . Discarding the remainder as lower order against  $c_B$  reduces (7) to the per-node condition  $|(E_L v^{(2)})_i| < \Delta\lambda |v_i^{(2)}|$  holding for every node  $i$ .

*Step 3 (concentration).* Lemma 3 bounds the left-hand side uniformly over the  $m$  nodes: with high probability,  $|(E_L v^{(2)})_i| \lesssim \sqrt{8\tilde{\rho}(\eta) \log m/p}$  with  $\tilde{\rho}(\eta) = (1 + \eta)\beta_0^2/\eta$ .

*Step 4 (synthesis).* The per-node condition of Step 2 is tightest on clan  $B$ , since  $\eta \geq 1$ . Squaring  $\sqrt{8\sigma_B^2 \log m} < \Delta\lambda c_B$  and substituting  $\sigma_B^2$ ,  $c_B^2 = 1/(m\eta)$  and the gap bound of Step 1, the factor  $\eta$  cancels and solving for  $p$  gives exactly (11) (Proposition 7 and corollary 5). Under (11) every node satisfies  $|w_i| < |v_i^{(2)}|$ , hence  $\hat{v}_i^{(2)}$  and  $v_i^{(2)}$  share a sign; since  $v^{(2)}$  is sign-constant on each clan (Lemma 4), the zero-threshold partition of  $\hat{v}^{(2)}$  is  $(A, B)$  exactly.  $\square$

Theorem 2 states the counterpart of Theorem 1 for the centered distance operator. Its proof follows the same four steps, with the population geometry of Step 1 and the variance load of Step 3 replaced by their distance analogues; Lemma 2 is applied unchanged, being a statement about an arbitrary symmetric perturbation of an arbitrary symmetric matrix. Section F carries the details.

## 5 Empirical Results

We test the recovery rate of Theorem 1 in two complementary regimes. In the *synthesized* regime, the similarity matrix  $S$  is built directly from the closed-form flat CBM (Assumption A3), isolating the linear algebra with no estimation noise. In the *generated* regime, dendropy simulates Kingman coalescent trees and evolves JC69 sequences of length  $\ell = 10^4$ , from which  $S$  is the log-det similarity (Assumption A1); this probes whether the CBM idealization survives realistic tree shapes and finite-sequence noise. Throughout, the rule that rounds the sampled Fiedler vector into a partition is  $k$ -means on the normalized Laplacian  $L_{\text{sym}}$ .

Recovery is scored by the Normalized Mutual Information (NMI) of the recovered split  $\hat{C}$  against the planted split  $C^*$ . NMI is the standard accuracy metric for evaluating spectral-clustering partitions against a known ground truth [14, 16]:

$$\text{NMI}(Y, C) = \frac{2I(Y; C)}{H(Y) + H(C)}, \quad (16)$$

with  $I(Y; C)$  the mutual information and  $H(\cdot)$  the Shannon entropy. NMI attains its maximum value 1 if and only if the two partitions are identical up to relabeling, and decays as they disagree [20, 14]. Composing this with the sign-recovery guarantee of Theorem 1 fixes the population endpoint of our sweeps.

**Corollary 4** (Perfect partition recovery and  $\text{NMI} = 1$ ). *Under the hypotheses of Theorem 1, the spectral partition  $\hat{C} = \{\hat{C}_A, \hat{C}_B\}$  obtained by thresholding  $\hat{v}^{(2)}$  at zero satisfies  $\hat{C} = C^*$  with high probability, and therefore  $\text{NMI}(\hat{C}, C^*) = 1$ .*

*Proof.* Theorem 1 guarantees  $\|\hat{v}^{(2)} - v^{(2)}\|_\infty < \min_i |v_i^{(2)}|$ , so for each node  $i$ ,  $|\hat{v}_i^{(2)} - v_i^{(2)}| < |v_i^{(2)}|$ , which forces  $\hat{v}_i^{(2)}$  and  $v_i^{(2)}$  to share a sign. Under the CBM the population Fiedler vector is sign-constant on each clan [2] ( $v_i^{(2)} > 0$  on  $A$ ,  $v_i^{(2)} < 0$  on  $B$ ), so the zero-threshold partition of  $\hat{v}^{(2)}$  recovers  $C^*$  exactly. Exact recovery gives  $I(\hat{C}; C^*) = H(\hat{C}) = H(C^*)$ , whence by (16)  $\text{NMI}(\hat{C}, C^*) = 1$ .  $\square$

For the synthesized regime, Figure 2 sweeps the non-balanced flat CBM (Assumption A3) at the same four imbalances used in the generated regime,  $\eta = 1, 5, 10, 15$  ( $S_{\text{in}} = 0.9$ ,  $S_{\text{out}}^{\text{max}} = 0.05$ ), across two decades of  $m$ . Within each cell the left panel shows the measured-NMI phase transition: the recovery curves sharpen and shift left as  $m$  grows, with the dashed 0.90 level marking the empirical threshold  $\hat{p}^*(m)$ . The right panel plots  $\hat{p}^*(m)$  against the  $\log m/m$  shape of Theorem 1's rate (11) on log-log axes. With no estimation noise present, the empirical threshold tracks that decay closely at every  $\eta$ .

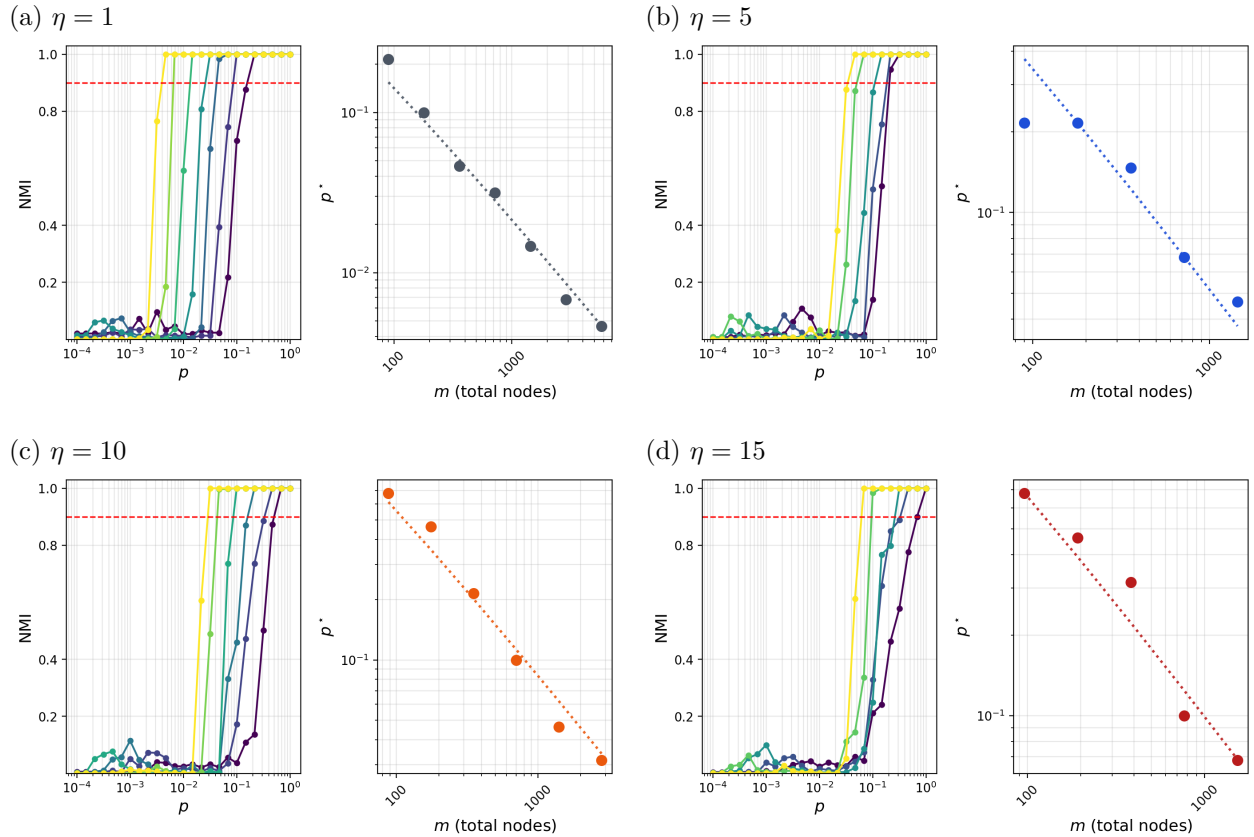


Figure 2: **Synthesized, non-balanced flat CBM** ( $S_{\text{in}} = 0.9$ ,  $S_{\text{out}}^{\text{max}} = 0.05$ ). One pair of panels per imbalance. Within each pair — left: recovery (measured NMI) vs sampling rate  $p$ , one curve per total size  $m$ , the curves running dark to light (viridis) with increasing  $m$  and the red dashed line at the 0.90 level that defines  $\hat{p}^*$ ; right:  $\hat{p}^*$  vs  $m$  on log-log axes (markers) against the dotted  $C \log m/m$  reference, the shape of (11) at fixed  $\eta$  with  $C$  measured as the median of the per-point ratios  $\hat{p}_i^*/(\log m_i/m_i)$ , giving  $C = \lceil \text{TBD} \rceil$  in (a)–(d). Each point averages 20 sub-sampling trials. No estimation noise is present:  $S$  is the closed-form block matrix.

Transitioning to the generated regime, we observe that on dendropy Kingman trees, the coherence and gap identities of Proposition 1 and lemma 1 continue to hold despite off-model tree shapes

and JC69 noise (Assumption A1). Specifically, the measured coherence  $\mu(U) \approx (1 + \eta)/2$ , Fiedler eigenvalue  $\lambda_2 \approx n S_{\text{out}}^{\max}$ , and spectral gap  $\Delta\lambda$  remains bounded above the theoretical lower bound  $m_{\min}(\rho - \eta S_{\text{out}}^{\max})$  (see Section G, Figure 7).

For the scaling behavior, Figure 3 sweeps an  $\eta$ -pooled set of Kingman samples at imbalances  $\eta \approx 1, 5, 10, 15$ . The left panel reports the measured NMI phase transition, and the right panel plots the empirical threshold  $\hat{p}^*$  (measured NMI  $\geq 0.95$ ) versus  $n$ . Within each row,  $\hat{p}^*(n)$  decays parallel to the dotted  $C \log n/n$  reference. This confirms that the  $\Theta(\log n/n)$  sample complexity predicted by (11) survives under realistic finite-sequence estimation noise.

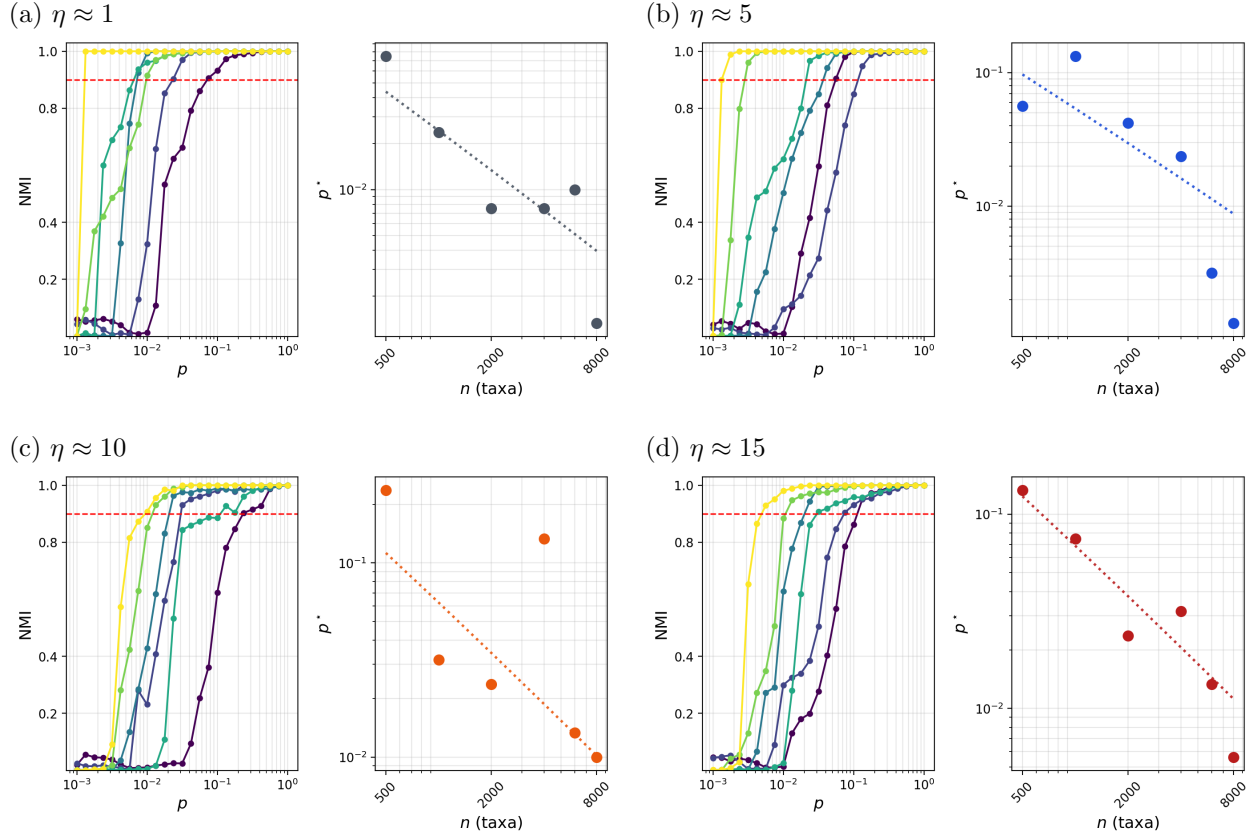


Figure 3: **Generated (Kingman),  $k$ -means rounding.** One pair of panels per imbalance. Within each pair — left: recovery (measured NMI) vs sampling rate  $p$ , one curve per  $n$ , the curves running dark to light (viridis) with increasing  $n = 500, 1000, 2000, 4000, 6000, 8000$  and the red dashed line at the 0.90 level that defines  $\hat{p}^*$ ; right:  $\hat{p}^*$  vs  $n$  on log-log axes (markers) against the dotted  $C \log n/n$  reference, the shape of (11) at fixed  $\eta$  with  $C$  measured as the median of the per-point ratios  $\hat{p}_i^*/(\log n_i/n_i)$ . Each point averages the pool samples available at that  $(n, \eta)$  — ten for  $\eta \approx 1, 5$ , three for  $\eta \approx 10, 15$ , and one at  $n = 8000$  throughout. Rounding is  $k$ -means on the normalized Laplacian  $L_{\text{sym}}$  applied to the sampled Fiedler vector.

## References

- [1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank. *Annals of Statistics*, 48(3):1452–1474, 2020.
- [2] Yariv Aizenbud, Ariel Jaffe, Meng Wang, Amber Hu, Noah Amsel, Boaz Nadler, Joseph T. Chang, and Yuval Kluger. Spectral top-down recovery of latent tree models. *Information and Inference: A Journal of the IMA*, 12(3):2300–2350, 2023.
- [3] David J. Aldous. Stochastic models and descriptive statistics for phylogenetic trees, from Yule to today. *Statistical Science*, 16(1):23–34, 2001.
- [4] Sivaraman Balakrishnan, Min Xu, Akshay Krishnamurthy, and Aarti Singh. Noise thresholds for spectral clustering. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 24, 2011.
- [5] Hans-Jürgen Bandelt and Andreas W. M. Dress. A canonical decomposition theory for metrics on a finite set. *Advances in Mathematics*, 92(1):47–105, 1992.
- [6] Michael G. B. Blum and Olivier François. Which random processes describe the tree of life? A large-scale study of phylogenetic tree imbalance. *Systematic Biology*, 55(4):685–691, 2006.
- [7] Emmanuel J. Candès and Yaniv Plan. Matrix completion with noise. *Proceedings of the IEEE*, 98(6):925–936, 2010.
- [8] Emmanuel J. Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.
- [9] Emmanuel J. Candès and Terence Tao. The power of convex relaxation: Near-optimal matrix completion. *IEEE Transactions on Information Theory*, 56(5):2053–2080, 2010.
- [10] Yudong Chen, Srinadh Bhojanapalli, Sujay Sanghavi, and Rachel Ward. Coherent matrix completion. In *Proceedings of the 31st International Conference on Machine Learning (ICML)*, pages 674–682. PMLR, 2014.
- [11] Justin Eldridge, Mikhail Belkin, and Yusu Wang. Unperturbed: spectral analysis beyond Davis-Kahan. In *Proceedings of Algorithmic Learning Theory (ALT)*, pages 321–358. PMLR, 2018.
- [12] Péter L. Erdős, Michael A. Steel, László A. Székely, and Tandy J. Warnow. A few logs suffice to build (almost) all trees (I). *Random Structures & Algorithms*, 14(2):153–184, 1999.
- [13] Jianqing Fan, Weichen Wang, and Yiqiao Zhong. An  $\ell_\infty$  eigenvector perturbation bound and its application to robust covariance estimation. *Journal of Machine Learning Research*, 18(207):1–42, 2018.
- [14] Santo Fortunato. Community detection in graphs. *Physics Reports*, 486(3–5):75–174, 2010.
- [15] Alexander Griffing. *Connections Between Numerical Taxonomy and Phylogenetics*. PhD thesis, North Carolina State University, 2012. Chapter 4: spectral structure of double-centred tree-additive distance matrices and partition recovery via the leading PCoA eigenvector.

- [16] Tai Qin and Karl Rohe. Regularized spectral clustering under the degree-corrected stochastic blockmodel. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 26, 2013.
- [17] Benjamin Recht. A simpler approach to matrix completion. *Journal of Machine Learning Research*, 12:3413–3430, 2011.
- [18] Charles Semple and Mike Steel. *Phylogenetics*, volume 24 of *Oxford Lecture Series in Mathematics and its Applications*. Oxford University Press, 2003.
- [19] Mike Steel and Andy McKenzie. Properties of phylogenetic trees generated by Yule-type speciation models. *Mathematical Biosciences*, 170(1):91–112, 2001.
- [20] Alexander Strehl and Joydeep Ghosh. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002.
- [21] Mark Wilkinson, James O. McInerney, Robert P. Hirt, Peter G. Foster, and T. Martin Embley. Of clades and clans: terms for phylogenetic relationships in unrooted trees. *Trends in Ecology & Evolution*, 22(3):114–115, 2007.

## A Proof of the spectral geometry

For the proof of Lemma 1 we first derive the spectral properties of the Fiedler vector.

**Lemma 4** (Spectral decomposition of the imbalanced Laplacian). *Under Assumptions A2 and A3: (i) the smallest eigenvalue is  $\lambda_1 = 0$  with  $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$ ; (ii) the Fiedler eigenvalue is  $\lambda_2 = m\beta_0$ ; (iii) the Fiedler vector is piecewise constant,  $v_i^{(2)} = +c_A$  for  $i \in A$  and  $v_i^{(2)} = -c_B$  for  $i \in B$ , with  $c_A = \sqrt{b/(am)}$  and  $c_B = \sqrt{a/(bm)}$ .*

*Proof.* For any  $L = D - S$ ,  $L\mathbf{1} = D\mathbf{1} - S\mathbf{1} = 0$ , so  $\lambda_1 = 0$  with  $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$ ; no assumption on  $S$  is needed.

*Step heights.* Assume  $v^{(2)}$  has the stated form with  $c_A, c_B > 0$ . The constraints  $\|v^{(2)}\|_2 = 1$  and  $\langle v^{(1)}, v^{(2)} \rangle = 0$  give  $ac_A = bc_B$ , hence

$$a\left(\frac{b}{a}c_B\right)^2 + bc_B^2 = 1 \implies \frac{bm}{a}c_B^2 = 1 \implies c_B^2 = \frac{a}{bm}, \quad c_A^2 = \frac{b}{am}.$$

*Eigenpair verification.* Write  $v^{(2)} = (c_A\mathbf{1}_a, -c_B\mathbf{1}_b)^\top$  with  $ac_A = bc_B$ . Using the block form of  $L$  and  $J_{a \times b}\mathbf{1}_b = b\mathbf{1}_a$ ,  $J_{b \times a}\mathbf{1}_a = a\mathbf{1}_b$ , together with  $S_A\mathbf{1}_a = \text{rowsum}(S_A)\mathbf{1}_a$ ,

$$Lv^{(2)} = \begin{pmatrix} (D_A - \text{rowsum}(S_A))c_A\mathbf{1}_a + b\beta_0c_B\mathbf{1}_a \\ -a\beta_0c_A\mathbf{1}_b - (D_B - \text{rowsum}(S_B))c_B\mathbf{1}_b \end{pmatrix}.$$

Since  $D_A = \text{rowsum}(S_A) + b\beta_0$  and  $D_B = \text{rowsum}(S_B) + a\beta_0$ , this reduces to  $(b\beta_0(c_A + c_B)\mathbf{1}_a, -a\beta_0(c_A + c_B)\mathbf{1}_b)^\top$ . From  $ac_A = bc_B$ ,  $b(c_A + c_B) = mc_A$  and  $a(c_A + c_B) = mc_B$ , so  $Lv^{(2)} = m\beta_0 v^{(2)}$  and  $\lambda_2 = m\beta_0$ .  $\square$

*Proof of Proposition 1.* Substituting the Fiedler coordinates of Lemma 4 into (4),

$$\mu(U) = \frac{m}{2} \max\left\{\frac{1}{m} + c_A^2, \frac{1}{m} + c_B^2\right\} = \frac{m}{2} \left(\frac{1}{m} + \frac{b}{ma}\right) = \frac{1}{2} \left(1 + \frac{b}{a}\right) = \frac{1+\eta}{2}. \quad \square$$

*Proof of Lemma 1.* The argument uses Assumptions A2, A3 and A5. By Lemma 4 the Fiedler eigenvalue is  $\lambda_2 = m\beta_0 = mS_{\text{out}}^{\max}$ . For  $\lambda_3$ , standard algebraic-connectivity interlacing gives  $\lambda_3(L) \geq \min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2}))$ , and applying the Fiedler-eigenvalue formula to each sub-clan Laplacian gives  $\lambda_2(L_{C_c}) \geq n_c S_{\text{in}}^{(c)}$ . With  $S_{\text{in}} = \min_c S_{\text{in}}^{(c)}$ ,

$$\min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2})) \geq \min(n_1, n_2) S_{\text{in}} = m_{\min} S_{\text{in}},$$

so  $\Delta\lambda = \lambda_3 - \lambda_2 \geq m_{\min} S_{\text{in}} - mS_{\text{out}}^{\max}$ . Using  $S_{\text{in}} = \rho + S_{\text{out}}^{\max}$  and  $m = m_{\min}(1 + \eta)$ ,

$$\Delta\lambda \geq m_{\min}(\rho + S_{\text{out}}^{\max}) - (1 + \eta)m_{\min}S_{\text{out}}^{\max} = m_{\min}\rho - \eta m_{\min}S_{\text{out}}^{\max}. \quad \square$$

*Proof of Corollary 1.* Positivity of the bound in Lemma 1 requires  $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$ . The minimum intra-clan similarity is bounded by the diameter,  $S_{\text{in}}^{\min} \geq e^{-r(\mathcal{T})}$ , and the maximum cross-clan similarity by the subtree depth,  $S_{\text{out}}^{\max} \propto e^{-\mathcal{O}(h(\mathcal{T}))}$ . Substituting into  $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$  gives (10); the logarithmic scaling  $r, h = \mathcal{O}(\log m)$  under Kingman yields  $\eta \leq \mathcal{O}(e^{\mathcal{O}(\log m)}) = \mathcal{O}(m^c)$ .  $\square$

## B Proof of the Laplacian sampling error

We bound  $\|E_L\|_2 \leq \|E_D\|_2 + \|E_S\|_2$ , where  $E_S = \hat{S} - S$  and  $E_D = \hat{D} - D$ , and show  $\|E_S\|_2$  dominates. We first record three structural facts about  $S$ .

**Proposition 2** (Rank-2 dominance of the rows). *Under Assumption A3,  $S$  is rank 2 and its row mass is captured by its leading rank-2 decomposition  $U\Lambda U^\top$ :  $\max_i \|S_{i,\cdot}\|_2 \leq \max_i \|e_i^\top U\Lambda U^\top\|_2$ , and  $\max_i \|e_i^\top U\Lambda U^\top\|_2^2 \leq \lambda_{\max}^2 \max_i \|e_i^\top U\|_2^2$ .*

*Proof.* The first inequality is immediate since  $S = U\Lambda U^\top$  under the CBM. For the second, put  $x^\top = e_i^\top U$ ; orthonormality of  $U$ 's columns preserves norm under  $U^\top$ , so with  $\Lambda = \text{diag}(\lambda_1, \lambda_2)$ ,  $\|x^\top \Lambda U^\top\|_2^2 = \lambda_1^2 x_1^2 + \lambda_2^2 x_2^2 \leq \lambda_{\max}^2 \|e_i^\top U\|_2^2$ .  $\square$

**Proposition 3** (Spectral scaling). *Under Assumptions A3 and A5,  $\lambda_{\max}(S) \leq m S_{\max}$ .*

*Proof.*  $S$  is non-negative, so by Perron–Frobenius its spectral radius is at most the maximum row sum,  $\lambda_{\max}(S) \leq \max_i \sum_j S_{ij} \leq m S_{\max}$ .  $\square$

**Proposition 4** (Row-norm / coherence bound). *Under Assumption A3,  $\max_i \|U_{i,\cdot}\|_2^2 = \frac{2\mu(U)}{m} = \frac{1+\eta}{m}$ .*

*Proof.* Immediate from (4) and  $\mu(U) = (1 + \eta)/2$  (Proposition 1).  $\square$

**Lemma 5** (Laplacian spectral error via matrix Bernstein). *Under Assumptions A3, A5 and A7, there are constants  $C, C_0$  such that if  $p \geq C_0 \frac{(1+\eta)\log m}{m}$  then with high probability*

$$\|E_L\|_2 \leq C \sqrt{\frac{m(1+\eta)\log m}{p}}.$$

*Proof. Bounding  $\|E_S\|_2$ .* Write  $E_S = \sum_{i \leq j} Z^{(ij)}$ , where  $Z^{(ij)}$  is supported on the symmetric pair  $(i, j)$  with value  $(\frac{1}{p}\Omega_{ij} - 1)S_{ij}$ ; the summands are independent and mean zero. The scale is  $\|Z^{(ij)}\|_2 \leq \max(\frac{1}{p} - 1, 1)S_{ij} \leq S_{\max}/p =: R$ . For the variance proxy,  $\mathbb{E}[(Z_{ij}^{(ij)})^2] = \frac{p(1-p)}{p^2} S_{ij}^2 \leq \frac{1}{p} S_{ij}^2$ , so  $\sigma^2 \leq \frac{1}{p} \max_i \|S_{i,\cdot}\|_2^2$ . Combining Propositions 2 to 4 with  $\max_i \|e_i^\top U\|_2^2 = 2\mu(U)/m$ ,

$$\max_i \|S_{i,\cdot}\|_2^2 \leq \lambda_{\max}^2 \frac{2\mu(U)}{m} \leq (m S_{\max})^2 \frac{2\mu(U)}{m} = 2m S_{\max}^2 \mu(U), \quad \sigma^2 \leq \frac{2}{p} m S_{\max}^2 \mu(U).$$

Matrix Bernstein,  $\mathbb{P}(\|E_S\|_2 \geq t) \leq 2m \exp(\frac{-t^2/2}{\sigma^2 + Rt/3})$ , gives in the variance-dominated regime, with probability  $1 - \epsilon$ ,  $\|E_S\|_2 \leq 2S_{\max} \sqrt{m \mu(U) \ln(2m/\epsilon)/p}$ .

*Bounding  $\|E_D\|_2$ .*  $E_D$  is diagonal with  $(E_D)_{ii} = \sum_j (\frac{1}{p}\Omega_{ij} - 1)S_{ij}$ ; a scalar Bernstein with  $\sigma_D^2 \leq \frac{1}{p} m S_{\max}^2$  and  $R_D = S_{\max}/p$  gives  $\|E_D\|_2 \lesssim S_{\max} \sqrt{m \ln(2m/\epsilon)/p}$ .

*Synthesis.* Since  $\|E_S\|_2 / \|E_D\|_2 = \Theta(\sqrt{\mu(U)}) \geq 1$ ,  $E_S$  dominates. Substituting  $\mu(U) = (1 + \eta)/2$  (Proposition 1),

$$\|E_L\|_2 \leq \|E_S\|_2 + \|E_D\|_2 \leq \tilde{C} \sqrt{\frac{m(1+\eta)\log m}{p}},$$

with  $\tilde{C}$  absorbing  $S_{\max}$  and absolute constants.  $\square$



## C Proof of the entry-wise perturbation

### C.1 Neumann decomposition (proof of Lemma 2)

Write  $\hat{L} = L + E_L$  and  $\hat{v}^{(2)} = v^{(2)} + w$ , and expand  $w$  into a resolvent power series in  $E_L$  [11].

**Lemma 6** (Exact Neumann decomposition). *Define*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad w^{(2)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L w \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)} + \langle v^{(2)}, w \rangle v^{(2)}.$$

Then  $w = w^{(1)} + w^{(2)}$  exactly, and under  $\|E_L\|_2 < \Delta\lambda$  this is the resolvent series  $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$ .

*Proof.* From  $\hat{L}\hat{v}^{(2)} = \hat{\lambda}_2 \hat{v}^{(2)}$  and  $Lv^{(2)} = \lambda_2 v^{(2)}$ ,  $(L - \hat{\lambda}_2 I)w = (\hat{\lambda}_2 - \lambda_2)v^{(2)} - E_L v^{(2)} - E_L w$ . Projecting onto  $v^{(\ell)}$  ( $\ell \neq 2$ ), using  $\langle v^{(\ell)}, v^{(2)} \rangle = 0$ , gives  $\langle v^{(\ell)}, w \rangle = (-\langle v^{(\ell)}, E_L v^{(2)} \rangle - \langle v^{(\ell)}, E_L w \rangle) / (\lambda_\ell - \hat{\lambda}_2)$ . Expanding  $w = \sum_\ell \langle v^{(\ell)}, w \rangle v^{(\ell)}$  and separating the  $\ell = 2$  component recovers the decomposition; iterating on  $E_L w$  under  $\|E_L\|_2 < \Delta\lambda$  yields the convergent series.  $\square$

**Lemma 7** (Entry-level first-order bound). *Provided  $\|E_L\|_2 < \Delta\lambda$ , for every node  $i$ ,*

$$|w_i^{(1)}| \leq \frac{|(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|}{\Delta\lambda - \|E_L\|_2}. \quad (17)$$

*Proof.* By Weyl's inequality  $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta\lambda - \|E_L\|_2$  for  $\ell \neq 2$ . Factoring the denominator,  $|w_i^{(1)}| \leq |(P_\perp E_L v^{(2)})_i| / (\Delta\lambda - \|E_L\|_2)$  with  $P_\perp = I - v^{(2)}(v^{(2)})^\top$ . Since  $P_\perp x = x - \langle v^{(2)}, x \rangle v^{(2)}$  and  $|\langle v^{(2)}, E_L v^{(2)} \rangle| \leq \|E_L\|_2 \|v^{(2)}\|_2^2 = \|E_L\|_2$ ,  $|(P_\perp E_L v^{(2)})_i| \leq |(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|$ .  $\square$

*Remark 4* (Asymmetric correction in the non-balanced CBM). The correction  $\|E_L\|_2 |v_i^{(2)}|$  equals  $\|E_L\|_2 c_A$  for  $i \in A$  and  $\|E_L\|_2 c_B$  for  $i \in B$ . For the binding clan  $B$  this is  $\|E_L\|_2 / \sqrt{m\eta}$ , strictly sub-dominant to the recovery target  $c_B$  when  $\|E_L\|_2 \ll \Delta\lambda$ . Throughout Section C.2 we therefore use the leading-order simplification  $|w_i^{(1)}| \lesssim |(E_L v^{(2)})_i| / \Delta\lambda$  for  $i \in B$ .

### C.2 Per-node Bernstein concentration

**Lemma 8** (Laplacian increment representation). *For every node  $i$ ,  $(E_L v^{(2)})_i = \sum_{j \neq i} X_{ij}$  with  $X_{ij} = (\frac{\omega_{ij}}{p} - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$  independent and mean zero.*

*Proof.* With  $(E_S)_{ij} = (\omega_{ij}/p - 1)S_{ij}$ ,  $(E_L)_{ii} = \sum_{k \neq i} (E_S)_{ik}$  and  $(E_L)_{ij} = -(E_S)_{ij}$ , so  $(E_L v^{(2)})_i = \sum_{k \neq i} (E_S)_{ik} v_i^{(2)} - \sum_{j \neq i} (E_S)_{ij} v_j^{(2)} = \sum_{j \neq i} (E_S)_{ij} (v_i^{(2)} - v_j^{(2)})$ .  $\square$

**Proposition 5** (Within-clan terms vanish). *Under Assumption A3,  $v^{(2)}$  is constant on each clan (Lemma 4), so  $X_{ij} = 0$  for same-clan pairs; only cross-clan pairs survive:  $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$  for  $i \in B$ , and  $(E_L v^{(2)})_i = \sum_{j \in B} X_{ij}$  for  $i \in A$ .*

**Proposition 6** (Per-node variance). *Under Assumptions A3, A5 and A7 with  $\beta_0 = S_{\text{out}}^{\max}$ , using  $(c_A + c_B)^2 = m/(ab) = (1 + \eta)^2/(m\eta)$ ,*

$$\sigma_B^2 := \text{Var}((E_L v^{(2)})_i)_{i \in B} = \frac{(1-p)(1+\eta)\beta_0^2}{\eta p}, \quad \sigma_A^2 = \frac{(1-p)(1+\eta)\beta_0^2}{p} = \eta \sigma_B^2, \quad (18)$$

so clan  $B$  has strictly smaller per-node variance;  $\sigma_B^2 = (1-p)\tilde{\rho}(\eta)/p$  with  $\tilde{\rho}(\eta) = (1+\eta)\beta_0^2/\eta$ .

*Proof.* For  $i \in B$  each of the  $a$  surviving terms has  $S_{ij} = \beta_0$  and  $|v_i^{(2)} - v_j^{(2)}| = c_A + c_B$ , so  $\sigma_B^2 = \frac{1-p}{p}(c_A + c_B)^2 a \beta_0^2$ ; substituting  $a = m/(1 + \eta)$  and  $(c_A + c_B)^2 = (1 + \eta)^2/(m\eta)$  gives the stated value. Clan  $A$  follows symmetrically with  $b$  in place of  $a$ .  $\square$

*Proof of Lemma 3.* By Lemma 8 and proposition 5, for  $i \in B$  the quantity  $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$  is a sum of independent, mean-zero summands with  $|X_{ij}| \leq \beta_0(c_A + c_B)/p = R_B$  and total variance  $\sigma_B^2$  (Proposition 6). Bernstein's inequality for bounded independent summands gives the stated two-term deviation bound at confidence  $1 - \delta$ . In the variance-dominated regime  $p = \Theta(\log m/m)$  the linear term is of lower order, and a union bound over the  $m$  nodes at  $\delta = \delta_0/m$  yields (15).  $\square$

**Proposition 7** (Clan  $B$  is binding). *Under Assumption A3,  $\min_i |v_i^{(2)}| = |c_B|$ , attained on the larger clan  $B$ . The per-node condition  $|(E_L v^{(2)})_i| < \Delta \lambda |v_i^{(2)}|$  requires  $p > 8m(1 + \eta)\beta_0^2 \log m / \Delta \lambda^2$  on  $B$  (the clan- $A$  condition carries an extra  $1/\eta$ ), so since  $\eta \geq 1$ , clan  $B$  binds.*

*Proof.* Squaring  $\sqrt{8\sigma_B^2 \log m} < \Delta \lambda c_B$  and substituting  $\sigma_B^2$  with  $c_B^2 = 1/(m\eta)$ : the  $\eta$  cancels, giving  $p > 8m(1 + \eta)\beta_0^2 \log m / \Delta \lambda^2$ . Clan  $A$  follows with  $c_A^2 = \eta/m$ .  $\square$

**Corollary 5** (Per-node recovery condition). *If  $|w_i| < \min_i |v_i^{(2)}|$  for all  $i$  then  $\text{sign}(\hat{v}^{(2)}) = \text{sign}(v^{(2)})$ . Combining Lemma 3 and proposition 7 with the gap bound  $\Delta \lambda \geq m(\rho - \eta\beta_0)/(1 + \eta)$  (Lemma 1) yields*

$$p > \frac{8(1 + \eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\max})^2} = \Theta\left(\frac{\log m}{m}\right), \quad (19)$$

*which is the rate (11) of Theorem 1. This computation is Step 4 of the proof in Section 4.*

### C.3 The second-order remainder

The bound below is the term discarded at Step 2 of Section 4; it is what restricts the argument to the strong-signal regime  $\|E_L\|_2 \ll \Delta \lambda$ .

**Lemma 9** (Second-order negligibility).  $\|w^{(2)}\|_\infty = O(\|E_L\|_2^2 / \Delta \lambda^2)$ .

*Proof.* By Davis–Kahan  $\|w\|_2 \leq \|E_L\|_2 / (\Delta \lambda - \|E_L\|_2)$ . Each coefficient  $b_\ell = -\langle v^{(\ell)}, E_L w \rangle$  satisfies  $|b_\ell| \leq \|E_L\|_2 \|w\|_2$ , and since  $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta \lambda - \|E_L\|_2$  each summand of  $w^{(2)}$  contributes  $O(\|E_L\|_2^2 / \Delta \lambda^2)$  in  $\ell_\infty$ ; the  $v^{(2)}$  component obeys  $|\langle v^{(2)}, w \rangle| = O(\|w\|_2^2) = O(\|E_L\|_2^2 / \Delta \lambda^2)$ .  $\square$

## D Extension: the Hierarchical Block Model

Here Assumption A3 is replaced by Assumption A4; Assumptions A1, A2 and A5 are retained. The CBM holds exactly under the strict clock with *symmetric* clans; for asymmetric (caterpillar-like) clans, within-clan MRCA depths vary even under the clock, so intra-clan entries are no longer constant. The HBM of Assumption A4 retains the constant cross-clan boundary but lets intra-clan similarities decay geometrically with path depth; its place in the chain of generative assumptions is recorded in Table 1. This appendix draws the two consequences: a geometric floor on the weakest intra-clan similarity, and the spectral-gap bound that floor induces.

**Proposition 8** (Geometric floor). *Under Assumption A4,  $S_{\text{in}}^{\min} \geq S_{\text{in}} \alpha^{D_{\max}}$  with  $D_{\max} = \lceil \log_2 \max(a, b) \rceil$ .*

*Proof.* GTR multiplicativity along paths gives  $S_{ij} = \prod_{e \in \text{path}(i,j)} \alpha_e$ , so a same-clan pair at depth  $\leq D_{\max}$  satisfies  $S_{ij} \geq S_{\text{in}} \alpha^{D_{\max}}$ ; minimising over same-clan pairs gives the floor, with  $D_{\max}$  the height of the deeper clan subtree.  $\square$

**Proposition 9** (HBM spectral-gap bound). *Substituting Proposition 8 into Lemma 1 gives*

$$\Delta\lambda \geq m_{\min}(S_{\text{in}}\alpha^{D_{\max}} - S_{\text{out}}^{\max}) - \eta m_{\min}S_{\text{out}}^{\max} = m_{\min}S_{\text{in}}\alpha^{D_{\max}} - (1 + \eta)m_{\min}S_{\text{out}}^{\max},$$

positive iff  $\alpha^{D_{\max}} > (1 + \eta)S_{\text{out}}^{\max}/S_{\text{in}}$ . As  $\alpha \rightarrow 1$  (or  $D_{\max} = 0$ ) this reduces to Lemma 1 and the CBM margin condition of Assumption A6.

Figure 4 verifies the bound on simulated HBM matrices, and Figure 5 illustrates the imbalance-driven degradation of the recovered partition.

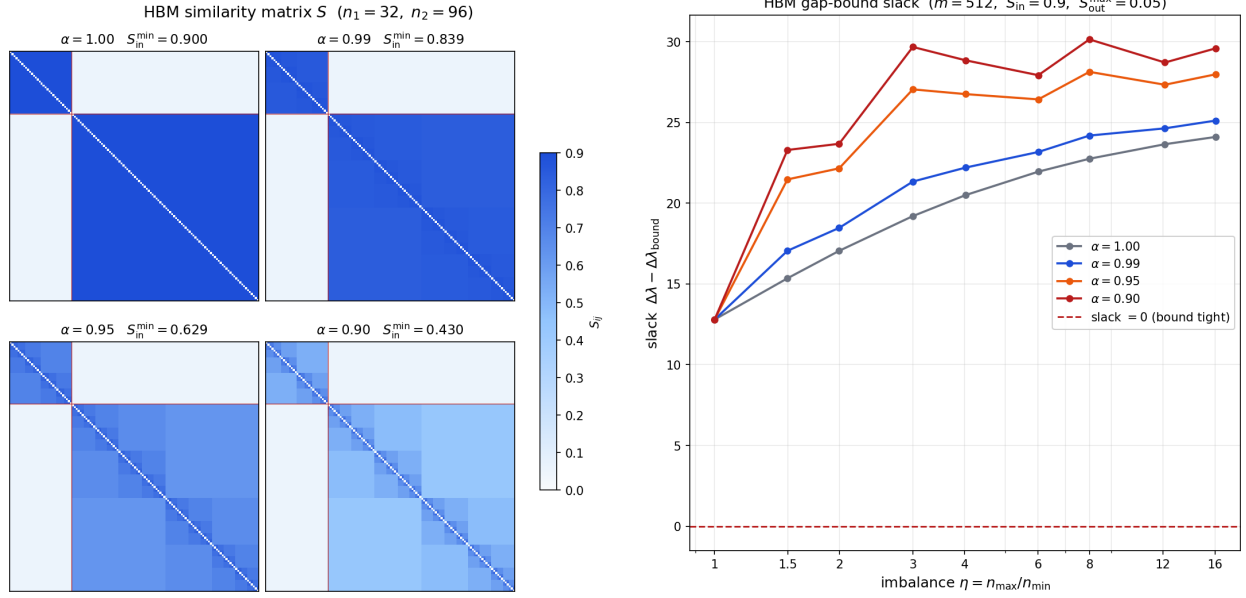


Figure 4: Empirical verification of Proposition 9 over a sweep of  $(\eta, m)$  across decay rates  $\alpha \in \{1.0, 0.99, 0.95, 0.90\}$ . **(Left)** HBM similarity matrices  $S$  degrading as  $\alpha$  decreases. **(Right)** slack of the gap lower bound versus  $\eta$ , resolving  $S_{\text{in}}^{\min} = S_{\text{in}}\alpha^{D_{\max}}$  at  $D_{\max} = \lceil \log_2 \max(n_1, n_2) \rceil$ ; the bound stays valid (slack  $\geq 0$ ) across all configurations.

## E The Neumann trick — comparison with standard bounds

The recovery criterion (7) constrains  $w = \hat{v}^{(2)} - v^{(2)}$  in the entry-wise sense,  $\|w\|_{\infty} < c_B$ , where  $c_B = \min_i |v_i^{(2)}|$  is the entry-wise margin, attained on the binding clan  $B$  with  $c_B^2 = 1/(m\eta)$ , while the sampled-Laplacian noise satisfies  $\|E_L\|_2 \leq \tilde{C}\sqrt{m(1+\eta)\log m/p}$  (Lemma 5). The two routes below bound the same  $w$ , differing only in how they relate the entry-wise constraint to  $E_L$ .

**The standard ( $\ell_2$ ) route.** Davis–Kahan controls only the Euclidean norm,  $\|w\|_2 \leq \|E_L\|_2/(\Delta\lambda - \|E_L\|_2)$ . Because the criterion is in the supremum norm, this must be transferred via  $\|w\|_{\infty} \leq \|w\|_2$ ; imposing  $\|w\|_2 < c_B$  and substituting the noise bound in the regime  $\|E_L\|_2 \ll \Delta\lambda$  yields  $p_{\text{naive}} \gtrsim m^2\eta(1+\eta)\log m/\Delta\lambda^2$ .

**The entry-wise (Neumann) route.** Lemma 2 bounds each coordinate directly, replacing  $\|E_L\|_2$  by the scalar  $(E_L v^{(2)})_i$  at the tightest coordinate  $i \in B$ . This scalar is a sum of  $O(m)$  independent

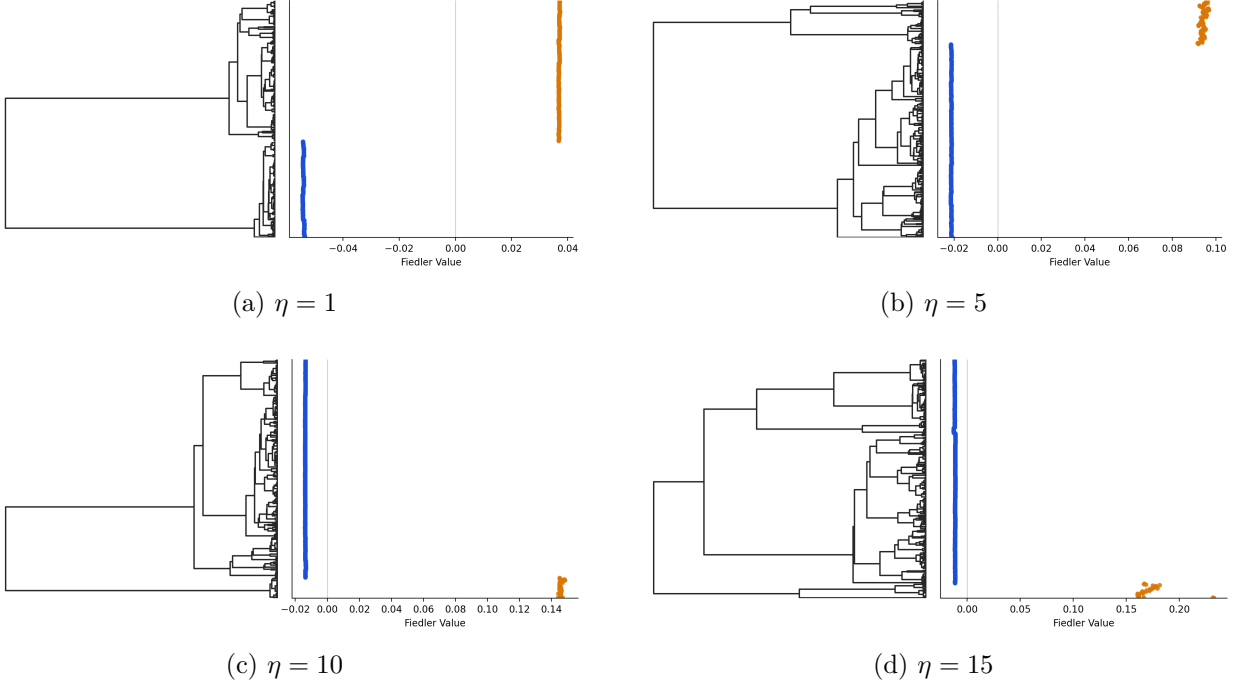


Figure 5: Fiedler bipartitions of a Kingman-model tree with 500 terminal nodes ( $n = 10,000$ ,  $d = 4$ , JC69), across increasing imbalance  $\eta$ .

mean-zero contributions concentrating at  $\sigma_B^2 = (1-p)(1+\eta)\beta_0^2/(\eta p)$  (Proposition 6), smaller than  $\|E_L\|_2^2$  by a factor  $m$ ; imposing  $|w_i| < c_B$  and the gap bound recovers the rate of Corollary 5.

*Remark 5* ( $\ell_2$  vs. entry-wise). Comparing the two rates gives  $p_{\text{naive}}/p_{\text{Neumann}} = \Theta(m\eta/\beta_0^2)$ , a  $\Theta(m)$  reduction for fixed  $\eta$ . The saving originates at the noise level: the operator norm  $\|E_L\|_2$  exceeds the individual increment  $(E_L v^{(2)})_i$  by  $\Theta(\sqrt{m})$ , and since  $p$  enters both rates as  $p^{-1}$  under a square root this squares to  $\Theta(m)$ . Both rates share  $(1+\eta)^3/(\rho - \eta\beta_0)^2$  and diverge as  $\rho - \eta\beta_0 \rightarrow 0$  (the boundary of Assumption A6), but the entry-wise rate stays within  $p \leq 1$  over a region where  $p_{\text{naive}}$  already exceeds it, and under the HBM decay of Proposition 9 remains feasible for larger  $D_{\text{max}}$ , accommodating deeper and more imbalanced trees than the Euclidean route admits.

## F The Distance Route

This appendix develops the distance route in full. Its structural basis is a single identity: for any tree-additive  $\mathcal{D}$ , the centred operator  $\mathcal{B} = HDH$  is a nonnegatively weighted sum of rank-one negative-semi-definite terms, one per edge (Lemma 10). Hence  $\mathcal{B} \preceq 0$ , its structural eigenvector belongs to the most negative eigenvalue, and each edge contributes at least  $2\tau_e n_1(e)n_2(e)/m$  to the magnitude of that eigenvalue (Lemma 11) — so the operator is by construction tuned to deep, balanced splits. Section F.1 builds  $\mathcal{D}$  and  $\mathcal{B}$  from data; Section F.2 proves the identity; Section F.3 states the sign rule; Section F.4 computes the population spectrum exactly in the balanced flat CBM; Section F.5 records what is known about which edge the rule selects on a general tree; Sections F.6 to F.8 carry the sub-sampling analysis through to Theorem 2. Two population inputs remain unresolved at general imbalance: a lower bound on  $\Delta\lambda^{(\mathcal{B})}$ , and the piecewise-constant shape of  $u^{(1)}$ . Both are carried as explicit hypotheses of Theorem 2 rather than absorbed silently.

## F.1 Building $\mathcal{D}$ and $\mathcal{B}$ from sequence data

Fix the tree  $\mathcal{T}$  on  $m$  leaves, the GTR rate matrix  $Q$  of Assumption A1, and the population similarity matrix  $S$  of Section 1. Throughout this appendix Assumption A2 is in force: it is what makes the transform below path-additive, since (1) expresses  $S_{ij}$  through the MRCA depth  $\tau_{ij}$ , and  $2\tau_{ij}$  is the path length between  $i$  and  $j$  only when the tree is ultrametric.

**Step 1 — distance from similarity.** Apply the entrywise log-transform

$$\mathcal{D}_{ij} = -\log S_{ij}, \quad \mathcal{D}_{ii} = 0. \quad (20)$$

Because  $S$  is tree-multiplicative,  $\mathcal{D}$  is then literally the tree-additive distance used elsewhere in this paper,  $\mathcal{D}_{ij} = \sum_{e \in \text{path}(i,j)} \tau_e$  with nonnegative edge lengths  $\tau_e$ .

**Step 2 — double centering.** Let  $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$  be the orthogonal projector onto  $\mathbf{1}^\perp$  ( $H^2 = H$ ,  $\|H\|_2 = 1$ ), and define

$$\mathcal{B} := H \mathcal{D} H = \mathcal{D} - \frac{1}{m} \mathbf{1}\mathbf{1}^\top \mathcal{D} - \frac{1}{m} \mathcal{D} \mathbf{1}\mathbf{1}^\top + \frac{1}{m^2} (\mathbf{1}^\top \mathcal{D} \mathbf{1}) \mathbf{1}\mathbf{1}^\top. \quad (21)$$

$H$  plays for  $\mathcal{B}$  the role that the row-sum-zero structure of  $L = D - S$  plays for the Laplacian: both operators annihilate  $\mathbf{1}$ ,  $L\mathbf{1} = 0$  and  $\mathcal{B}\mathbf{1} = 0$ . Computationally,  $\mathcal{B}$  is formed from  $\mathcal{D}$  in  $O(m^2)$  time (row means, column means, grand mean, four broadcasts).

## F.2 The split-decomposition identity

The following identity makes  $\mathcal{B} \preceq 0$  rigorous. It is used twice: for the sign convention of Section F.3 and in Section F.5. Binarity plays no role in it, so it is stated for an arbitrary tree.

**Lemma 10** (Split decomposition of  $\mathcal{B}$ , after Bandelt–Dress [5]). *Let  $\mathcal{T}$  be a tree with leaf set of size  $m$ , nonnegative edge lengths  $\{\tau_e\}_{e \in E(\mathcal{T})}$ , and tree-additive leaf distance  $\mathcal{D}_{ij} = \sum_{e \in \text{path}(i,j)} \tau_e$ . For each edge  $e$  let  $s_e \in \{-1, +1\}^m$  be the signed split indicator of the leaf bipartition  $e$  induces, and  $\delta_e := \frac{1}{2}(\mathbf{1}\mathbf{1}^\top - s_e s_e^\top)$ . Since  $(s_e)_i^2 = 1$  for every  $i$ ,  $\delta_e$  automatically has zero diagonal; moreover  $(\delta_e)_{ij} = 1$  iff  $i, j$  lie on opposite sides of  $e$ . Then*

$$\mathcal{D} = \sum_{e \in E(\mathcal{T})} \tau_e \delta_e, \quad \mathcal{B} = H \mathcal{D} H = -\frac{1}{2} \sum_{e \in E(\mathcal{T})} \tau_e (H s_e)(H s_e)^\top, \quad (22)$$

so  $\mathcal{B}$  is a nonnegatively weighted sum of rank-one negative-semi-definite terms, hence  $\mathcal{B} \preceq 0$ . For an edge separating  $n_1(e)$  leaves from  $n_2(e) = m - n_1(e)$ ,

$$\|H s_e\|_2^2 = m - \frac{(n_1 - n_2)^2}{m} = \frac{4 n_1(e) n_2(e)}{m}. \quad (23)$$

*Proof.*  $(\delta_e)_{ij} = \frac{1}{2}(1 - (s_e)_i(s_e)_j)$  equals 1 when  $(s_e)_i \neq (s_e)_j$  and 0 when they agree, giving the cut-metric claim; summing  $\tau_e \delta_e$  over edges and matching path membership against split membership gives the first identity in (22) termwise. Centering,  $H \mathbf{1}\mathbf{1}^\top H = 0$  since  $H\mathbf{1} = 0$ , so  $H \delta_e H = -\frac{1}{2}(H s_e)(H s_e)^\top$ , which summed over  $e$  gives the second identity. Each summand is negative-semi-definite (rank-one with a nonpositive coefficient), and a nonnegative combination of NSD matrices is NSD. The norm identity is the standard centered-indicator computation,  $\mathbf{1}^\top s_e = n_1 - n_2$  and  $\|s_e\|_2^2 = m$ .  $\square$

**Lemma 11** (Edge lower bound on the structural eigenvalue). *For every edge  $e$  of  $\mathcal{T}$ ,*

$$|\lambda_1(\mathcal{B})| \geq \frac{2\tau_e n_1(e) n_2(e)}{m}. \quad (24)$$

*Proof.* Put  $v = Hs_e / \|Hs_e\|_2$  in the Rayleigh quotient of  $-\mathcal{B} \succeq 0$ . By (22) every term of  $v^\top(-\mathcal{B})v$  is nonnegative, so retaining only the  $e$ -term gives  $v^\top(-\mathcal{B})v \geq \frac{1}{2}\tau_e \|Hs_e\|_2^2$ , and (23) evaluates the norm.  $\square$

### F.3 Using $\mathcal{B}$ : the sign-recovery rule

By Lemma 10,  $\mathcal{B} \preceq 0$  for any tree-additive  $\mathcal{D}$ , so the eigenvalue of largest magnitude,  $\lambda_1(\mathcal{B})$ , is the most negative one; let  $u^{(1)}$  denote a corresponding unit eigenvector. The sign pattern of  $u^{(1)}$  defines a bipartition,

$$\hat{A} = \{i : u_i^{(1)} \geq 0\}, \quad \hat{B} = \{i : u_i^{(1)} < 0\}. \quad (25)$$

That this bipartition is the structural split  $\{A, B\}$  is a property of the population operator, not a consequence of Lemma 10: it holds in the balanced flat CBM, where Proposition 10 exhibits  $u^{(1)} = (e_A - e_B)/\sqrt{m}$ , and on general trees it is the content of the result reported in Section F.5. Where it holds, and given an estimator  $\hat{u}^{(1)}$  built from a sub-sampled  $\mathcal{D}$  (Section F.6), a sufficient condition for  $\{\hat{A}, \hat{B}\} = \{A, B\}$  — as an unordered pair, since  $u^{(1)}$  is determined only up to a global sign — is the exact counterpart of (7),

$$\left\| \hat{u}^{(1)} - u^{(1)} \right\|_\infty < \min_i |u_i^{(1)}|, \quad (26)$$

the sign of  $\hat{u}^{(1)}$  being fixed by  $\langle \hat{u}^{(1)}, u^{(1)} \rangle \geq 0$ . Structurally (26) is identical to (7); only the operator producing the target vector changes. This is the criterion Theorem 2 certifies.

### F.4 Population-level landmarks

Under Assumption A3 with clans  $A, B$ , sizes  $a = |A| \leq b = |B|$ , imbalance  $\eta = b/a$ , and molecular-clock baseline  $\beta_0$  (Assumption A2), three population quantities feed Theorem 2:

1. **Cross-clan distance scale.** Every cross-clan entry of  $\mathcal{D}$  collapses to  $d_0 := -\log \beta_0$ , and  $d_0 > \mathcal{D}_{ij}$  for every within-clan pair; hence  $\max_{ij} \mathcal{D}_{ij} = d_0$ .
2. **Tree diameter.** By the previous item the diameter  $r(\mathcal{T}) = \max_{i,j} \mathcal{D}_{ij}$  equals  $d_0$  exactly under Assumption A2, so the diameter contributes no penalty separate from  $d_0$  and must not be charged twice (Section F.8).
3. **Top spectral gap of  $\mathcal{B}$ .** Since  $u^{(1)}$  belongs to the most negative eigenvalue and  $\mathcal{B} \preceq 0$ , the eigenvalue nearest  $\lambda_1(\mathcal{B})$  is the one of second-largest magnitude, so the separation relevant to Lemma 7 is  $\Delta\lambda^{(\mathcal{B})} := |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})|$ .

**Proposition 10** (Spectrum of  $\mathcal{B}$  under balanced flat CBM). *Assume the molecular-clock flat CBM with  $|A| = |B| = m/2$ , constant within-clan distance  $d_1$ , and constant cross-clan distance  $d_0 > d_1$ . Then*

$$\sigma(\mathcal{D}) = \left\{ \frac{(m-2)d_1 + md_0}{2}, \frac{(m-2)d_1 - md_0}{2}, \underbrace{-d_1, \dots, -d_1}_{m-2} \right\}, \quad (27)$$

$$\sigma(\mathcal{B}) = \left\{ 0, -\frac{m(d_0 - d_1)}{2} - d_1, \underbrace{-d_1, \dots, -d_1}_{m-2} \right\}. \quad (28)$$

The structural eigenvalue is  $\lambda_1(\mathcal{B}) = -\frac{m(d_0-d_1)}{2} - d_1$ , with eigenvector  $(e_A - e_B)/\sqrt{m}$ ; the next spectral cluster sits at  $-d_1$  with multiplicity  $m - 2$ ; and

$$\Delta\lambda^{(\mathcal{B})} = |\lambda_1(\mathcal{B})| - |\lambda_2(\mathcal{B})| = \frac{m(d_0 - d_1)}{2}. \quad (29)$$

*Proof.* Let  $e_A, e_B$  be the clan indicator vectors,  $\mathbf{1} = e_A + e_B$ , and write  $\mathcal{D} = T - d_1 I$  with  $T = d_1(e_A e_A^\top + e_B e_B^\top) + d_0(e_A e_B^\top + e_B e_A^\top)$ , which has  $T_{ij} = d_1$  within a clan (including  $i = j$ ) and  $T_{ij} = d_0$  across clans.  $T$  has rank 2 on  $\text{span}\{e_A, e_B\}$ ; in the orthonormal basis  $\hat{e}_A = e_A/\sqrt{m/2}$ ,  $\hat{e}_B = e_B/\sqrt{m/2}$ ,  $T|_{\text{span}\{\hat{e}_A, \hat{e}_B\}} \cong \frac{m}{2} \begin{pmatrix} d_1 & d_0 \\ d_0 & d_1 \end{pmatrix}$ , with eigenpairs  $(\frac{m(d_1+d_0)}{2}, \frac{1}{\sqrt{m}}\mathbf{1})$  and  $(\frac{m(d_1-d_0)}{2}, \frac{1}{\sqrt{m}}(e_A - e_B))$ ; the remaining  $m - 2$  directions lie in  $\ker T$ . Subtracting  $d_1 I$  shifts every eigenvalue by  $-d_1$ , giving  $\sigma(\mathcal{D})$ . For  $\mathcal{B} = H\mathcal{D}H$ : the eigenvector  $\mathbf{1}/\sqrt{m}$  lies in  $\ker H$  and contributes eigenvalue 0; the remaining  $m - 1$  eigenvectors of  $\mathcal{D}$  are already orthogonal to  $\mathbf{1}$  (the structural direction because  $|A| = |B|$ , the bulk by construction), so  $H$  fixes them and they remain eigenvectors of  $\mathcal{B}$  at the same eigenvalue. Reading off the extremes gives  $|\lambda_1(\mathcal{B})| = \frac{m(d_0-d_1)}{2} + d_1$  and  $|\lambda_2(\mathcal{B})| = d_1$ , hence (29).  $\square$

*Remark 6* (Realization of the flat CBM). Constant within-clan distance forces every within-clan pair to have the same MRCA, so the flat CBM is realized by two stars joined by an edge rather than by a binary tree. It is used here only as the one configuration in which the spectrum of  $\mathcal{B}$  can be computed in closed form; Lemmas 10 and 11 hold for arbitrary trees and are what the general-tree statements rest on.

*Remark 7* (Unbalanced case: open). Proposition 10 does not extend to  $a \neq b$  by the same argument. The proof uses that the structural eigenvector of  $\mathcal{D}$  is orthogonal to  $\mathbf{1}$ , so that  $H$  acts on it as the identity; that orthogonality is exactly what fails when  $a \neq b$ . Consequently neither a closed form for  $\Delta\lambda^{(\mathcal{B})}$  nor the piecewise-constant shape of  $u^{(1)}$  is available at general  $\eta$ , which is why both are hypotheses of Theorem 2. Lemma 11 does bound  $|\lambda_1(\mathcal{B})|$  from below at any imbalance, but bounding  $|\lambda_2(\mathcal{B})|$  from above — and hence the gap from below — remains open.

## F.5 Which edge the sign rule selects

Proposition 10 is exact under the flat-CBM idealisation. On an arbitrary tree, Griffing [15, Ch. 4] shows that, under a separation hypothesis on the maximiser of  $\tau_e n_1(e) n_2(e)$ , the sign partition of  $u^{(1)}$  is the leaf bipartition obtained by deleting that edge, so  $\hat{A}, \hat{B}$  are adjacent clans of  $\mathcal{T}$ . We state this as a citation rather than as a numbered result: the separation hypothesis is quantitative in the source and we do not reproduce it, and nothing in Theorem 2 depends on it — its only population inputs are  $\Delta\lambda^{(\mathcal{B})} > 0$  and the shape of  $u^{(1)}$ .

What this paper proves in the same direction is Lemma 11: every edge lower-bounds  $|\lambda_1(\mathcal{B})|$  by  $2\tau_e n_1(e) n_2(e)/m$ , so  $\tau_e n_1(e) n_2(e)$  is the correct objective to associate with the leading eigenvalue, independently of whether its maximiser’s eigenvector converges to  $Hs_{e^*}$ .

*Remark 8* (Which edge each route targets). The two routes can select different edges of the same tree. By Lemma 11 the weight an edge contributes to  $|\lambda_1(\mathcal{B})|$  is proportional to  $\tau_e n_1(e) n_2(e)$ , which is maximised by deep and *balanced* edges; the Fiedler vector of  $L$ , by contrast, is governed by algebraic connectivity and can prefer a fringe edge when within-clan variation inflates spurious low-Rayleigh directions [2, Thm. 4.2]. This is a statement about which split each rule targets at  $p = 1$ , and it is independent of the noise-tolerance comparison of Section F.8: the two answer different questions, and only the latter is settled by the rates.

## F.6 Sampling model and noise operator

Sample each off-diagonal upper-triangular entry of  $\mathcal{D}$  independently with probability  $p \in (0, 1]$ , exactly as in Assumption A7, and rescale by inverse probability,

$$\hat{\mathcal{D}}_{ij} = \frac{\Omega_{ij}}{p} \mathcal{D}_{ij}, \quad \mathbb{E}[\hat{\mathcal{D}}_{ij}] = \mathcal{D}_{ij}. \quad (30)$$

Center the noisy distances,  $\hat{\mathcal{B}} := H\hat{\mathcal{D}}H$ , and write  $E_{\mathcal{B}} := \hat{\mathcal{B}} - \mathcal{B} = H E_{\mathcal{D}} H$  with  $E_{\mathcal{D}} := \hat{\mathcal{D}} - \mathcal{D}$  the uncentered entrywise error. Idempotence of  $H$  and  $\|H\|_2 = 1$  immediately give

$$\|E_{\mathcal{B}}\|_2 \leq \|E_{\mathcal{D}}\|_2. \quad (31)$$

The entries of  $E_{\mathcal{D}}$  are bounded by  $\max_{ij} \mathcal{D}_{ij}/p = d_0/p$  (Section F.4); the diameter therefore enters only through  $d_0$ . The Neumann machinery of Section C.1 requires an operator-norm bound, which the following supplies; it is the distance analogue of Lemma 5 and is proved the same way.

**Lemma 12** (Distance spectral error via matrix Bernstein). *Under Assumptions A2, A3 and A7 there are constants  $C_1, C_0 > 0$  such that if  $p \geq C_0 \log m/m$  then, with high probability,*

$$\|E_{\mathcal{B}}\|_2 \leq \|E_{\mathcal{D}}\|_2 \leq C_1 d_0 \sqrt{\frac{m \log m}{p}}. \quad (32)$$

*Proof.* Write  $E_{\mathcal{D}} = \sum_{i < j} Z^{(ij)}$  with  $Z^{(ij)}$  supported on the symmetric pair  $(i, j)$  and value  $(\frac{1}{p}\Omega_{ij} - 1)\mathcal{D}_{ij}$ ; the summands are independent and mean zero, with scale  $\|Z^{(ij)}\|_2 \leq \mathcal{D}_{ij}/p \leq d_0/p =: R$  and  $\mathbb{E}[(Z^{(ij)})^2] \leq \mathcal{D}_{ij}^2/p$ . Hence the variance proxy satisfies  $\sigma^2 \leq \frac{1}{p} \max_i \|\mathcal{D}_{i,\cdot}\|_2^2 \leq m d_0^2/p$ . Matrix Bernstein gives  $\|E_{\mathcal{D}}\|_2 \lesssim \sqrt{\sigma^2 \log m} + R \log m = d_0 \sqrt{m \log m/p} + d_0 \log m/p$ , and the first term dominates once  $p \geq C_0 \log m/m$ . Equation (31) transfers the bound to  $E_{\mathcal{B}}$ .  $\square$

## F.7 Per-node concentration

The Neumann/resolvent machinery that turns control of  $E_{\mathcal{B}}$  into coordinate-wise control of  $\hat{u}^{(1)} - u^{(1)}$  (Lemmas 6, 7 and 9) is a statement about an arbitrary symmetric perturbation of an arbitrary symmetric matrix, so it transfers unchanged under  $L \rightarrow \mathcal{B}$ ,  $v^{(2)} \rightarrow u^{(1)}$ ,  $\Delta\lambda \rightarrow \Delta\lambda^{(\mathcal{B})}$ , given the operator-norm bound of Lemma 12. The concentration step of Section C.2 does not transfer: Lemma 8 writes  $(E_L v^{(2)})_i = \sum_{j \neq i} X_{ij}$  with  $X_{ij} = (\omega_{ij}/p - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$ , a Laplacian row-sum identity, and that is what lets Proposition 5 kill every within-clan increment. Since  $\mathcal{B}$  has no row-sum-zero structure on individual rows — only  $\mathcal{B}\mathbf{1} = 0$  globally, through  $H$  — no such cancellation is available here. None is needed: the increments weight  $u_j^{(1)}$  rather than a difference, and  $u^{(1)}$  is a unit vector, which bounds the whole sum at once.

Since  $u^{(1)} \perp \mathbf{1}$  we have  $Hu^{(1)} = u^{(1)}$ , hence

$$(E_{\mathcal{B}}u^{(1)})_i = \underbrace{(E_{\mathcal{D}}u^{(1)})_i}_{\text{per-node sum}} - \underbrace{\frac{1}{m}\mathbf{1}^\top E_{\mathcal{D}}u^{(1)}}_{\text{global centering scalar}}. \quad (33)$$

**Lemma 13** (Per-node variance and scale, no cancellation required). *Under Assumptions A2, A3 and A7, for every node  $i$ ,*

$$\text{Var}((E_{\mathcal{D}}u^{(1)})_i) = \frac{1-p}{p} \sum_{j \neq i} \mathcal{D}_{ij}^2 |u_j^{(1)}|^2 \leq \frac{d_0^2}{p}, \quad (34)$$



and each summand of  $(E_{\mathcal{D}}u^{(1)})_i$  is bounded in magnitude by  $R := d_0 \max_j |u_j^{(1)}|/p$ . If in addition  $u^{(1)}$  is piecewise constant on  $(A, B)$  with  $\|u^{(1)}\|_2 = 1$ , then  $\max_j |u_j^{(1)}| = \sqrt{\eta/m}$  and  $\min_j |u_j^{(1)}| = 1/\sqrt{m\eta}$ .

*Proof.* The summands  $(E_{\mathcal{D}})_{ij}u_j^{(1)}$  are independent and mean zero with variance  $\frac{1-p}{p}\mathcal{D}_{ij}^2|u_j^{(1)}|^2$ , giving the first equality. Bounding  $\mathcal{D}_{ij} \leq d_0$  (Section F.4) and using  $\sum_j |u_j^{(1)}|^2 = 1$  gives  $\leq d_0^2/p$ ; no within-clan cancellation is invoked. The scale bound is immediate from  $|(E_{\mathcal{D}})_{ij}| \leq d_0/p$ . For the last claim,  $u^{(1)} \perp \mathbf{1}$ , because  $\mathcal{B}\mathbf{1} = 0$  and  $\lambda_1(\mathcal{B}) \neq 0$ , so the two constants have opposite signs and satisfy  $a c_A = b c_B$ ; with  $a c_A^2 + b c_B^2 = 1$  and  $b/a = \eta$  this forces  $c_A = \sqrt{\eta/m}$  on  $A$  and  $-c_B = -1/\sqrt{m\eta}$  on  $B$ .  $\square$

**Lemma 14** (Per-node concentration for  $\mathcal{B}$ ). *Assume Assumptions A2, A3 and A7 and let  $u^{(1)}$  be piecewise constant on  $(A, B)$ . Then, with high probability uniformly in  $i$ ,*

$$|(E_{\mathcal{D}}u^{(1)})_i| \lesssim d_0 \sqrt{\frac{\log m}{p}} + \frac{d_0 \log m}{p} \sqrt{\frac{\eta}{m}}, \quad (35)$$

and the second term is dominated by the first as soon as  $p \geq \eta \log m/m$ .

*Proof.* Apply Bernstein's inequality to the independent mean-zero summands of  $(E_{\mathcal{D}}u^{(1)})_i$  with variance proxy  $d_0^2/p$  and scale  $R = d_0 \max_j |u_j^{(1)}|/p = d_0 \sqrt{\eta/m}/p$  (Lemma 13), at confidence  $1 - \delta_0/m$ , and take a union bound over the  $m$  nodes. Comparing the two terms gives the stated crossover.  $\square$

*Remark 9* (The centering scalar). The second term of (33) is a scalar, identical at every node, and it is carried in the budget rather than discarded. Cauchy-Schwarz and Lemma 12 give  $|\frac{1}{m}\mathbf{1}^\top E_{\mathcal{D}}u^{(1)}| \leq \|E_{\mathcal{D}}\|_2/\sqrt{m} \lesssim d_0 \sqrt{\log m/p}$ , the same order as the per-node term of Lemma 14, so

$$|(E_{\mathcal{B}}u^{(1)})_i| \leq |(E_{\mathcal{D}}u^{(1)})_i| + \frac{\|E_{\mathcal{D}}\|_2}{\sqrt{m}} \lesssim d_0 \sqrt{\frac{\log m}{p}} : \quad (36)$$

it costs a constant factor, not a factor of  $m$ . Note that it cannot be removed by de-meaning, since  $\hat{\mathcal{B}}\mathbf{1} = 0$  already forces  $\hat{u}^{(1)}$  to have mean zero.

## F.8 Proof of Theorem 2 and comparison

*Proof of Theorem 2.* The argument is the synthesis of Section 4 with  $L \rightarrow \mathcal{B}$ ,  $v^{(2)} \rightarrow u^{(1)}$ ,  $\Delta\lambda \rightarrow \Delta\lambda^{(\mathcal{B})}$ .

*Step 1 (population geometry).* By hypothesis (ii) and Lemma 13 the entry-wise margin is  $\min_i |u_i^{(1)}| = 1/\sqrt{m\eta}$ , attained on  $B$ , and  $\Delta\lambda^{(\mathcal{B})} > 0$  by (i). Hypothesis (iii) supplies  $\|E_{\mathcal{B}}\|_2 \leq \varepsilon_m \Delta\lambda^{(\mathcal{B})}$  with  $\varepsilon_m \leq c_0(m\eta)^{-1/4}$ , which in particular puts the perturbation in the strong-signal regime  $\|E_{\mathcal{B}}\|_2 < \Delta\lambda^{(\mathcal{B})}$  required by Lemma 6. Since  $\Delta\lambda^{(\mathcal{B})} \leq |\lambda_1(\mathcal{B})| \leq m d_0$ , (12) implies  $p > C\eta \log m/m$ , which discharges both the crossover condition of Lemma 14 and the hypothesis of Lemma 12 for  $C \geq \max(1, C_0)$ .

*Step 2 (entry-wise reduction).* Write  $\hat{u}^{(1)} = u^{(1)} + w$  and  $w = w^{(1)} + w^{(2)}$  as in Lemma 6. By Lemma 7,  $|w_i^{(1)}| \leq (|(E_{\mathcal{B}}u^{(1)})_i| + \|E_{\mathcal{B}}\|_2 |u_i^{(1)}|)/(\Delta\lambda^{(\mathcal{B})} - \|E_{\mathcal{B}}\|_2)$ , and by Lemma 9  $\|w^{(2)}\|_\infty = O(\|E_{\mathcal{B}}\|_2^2/(\Delta\lambda^{(\mathcal{B})})^2) = O(\varepsilon_m^2)$ . Hypothesis (iii) is what makes both corrections lower order against the margin:  $\varepsilon_m^2 \leq c_0^2(m\eta)^{-1/2} = o(\min_i |u_i^{(1)}|)$  and  $\|E_{\mathcal{B}}\|_2 |u_i^{(1)}|/(\Delta\lambda^{(\mathcal{B})} - \|E_{\mathcal{B}}\|_2) \leq \frac{\varepsilon_m}{1-\varepsilon_m} |u_i^{(1)}|$ ,

a vanishing fraction of the margin. Discarding them reduces (26) to the per-node condition  $|(E_B u^{(1)})_i| < \frac{1}{2} \Delta \lambda^{(B)} |u_i^{(1)}|$  for every  $i$ , and (36) bounds its left-hand side by a constant multiple of  $d_0 \sqrt{\log m/p}$ .

*Step 3 (concentration).* Lemma 14 with (36) bounds that left-hand side by  $C_2 d_0 \sqrt{\log m/p}$  uniformly over the  $m$  nodes, with high probability.

*Step 4 (synthesis).* The per-node condition is tightest on clan  $B$ , where the margin  $1/\sqrt{m\eta}$  is smallest. Imposing  $C_2 d_0 \sqrt{\log m/p} < \frac{1}{2} \Delta \lambda^{(B)} / \sqrt{m\eta}$  and solving for  $p$  gives (12) with  $C = 4C_2^2$ . The failure probability is the union bound of Step 3 together with Lemma 12, i.e.  $O(m^{-c})$  for some  $c > 0$ .  $\square$

*Remark 10* (Two normalizations, not to be mixed). Equation (12) carries  $m$  in the numerator and the unsubstituted  $(\Delta \lambda^{(B)})^2$  in the denominator. The similarity-side rate (11) looks different only because one further substitution has been made there:  $\Delta \lambda \geq m_{\min}(\rho - \eta S_{\text{out}}^{\max})$  with  $m_{\min} = m/(1 + \eta)$  (Lemma 1) turns  $m \log m / \Delta \lambda^2$  into  $(1 + \eta)^2 \log m / [m(\rho - \eta S_{\text{out}}^{\max})^2]$ , which is where the  $(1 + \eta)^3$  and the  $m$  in the denominator come from. Applying that conversion while retaining  $(\Delta \lambda^{(B)})^2$  would count it twice. The distance route stops before it because no lower bound on  $\Delta \lambda^{(B)}$  is available at general  $\eta$  (Remark 7); Corollary 2 performs the substitution in the one case where the gap is known.

**Comparison with the Laplacian route.** Substituting the balanced-case gap of Proposition 10 gives Corollary 2, and with it the only regime in which the two rates are directly comparable. There they share the exponent  $\Theta(\log m/m)$  and differ in one quantity: the cross-clan noise scale  $\beta_0$  is replaced by  $d_0 = -\log \beta_0$ , so for the same target split the distance route pays

$$\frac{d_0^2 (\rho - \eta S_{\text{out}}^{\max})^2}{\beta_0^2 (d_0 - d_1)^2}, \quad (37)$$

evaluated at  $\eta = 1$ , the only imbalance for which Corollary 2 applies. At the synthesized-regime values of Section 5 ( $\beta_0 = 0.05$ ,  $S_{\text{in}} = 0.9$ , hence  $d_0 = 2.996$ ,  $d_1 = 0.105$ ) the factor is 275. This compares the two parameter groups, not the two stated right-hand sides: (11) carries the explicit  $8(1 + \eta)^3$  while (13) carries an unspecified  $4C$ , and by Remark 1 neither constant is currently pinned down. Two caveats attach to it. First, the feasibility threshold  $\eta = \Omega((m/\log m)^{1/3})$  of Corollary 3 does *not* transfer to the distance route: it is extracted from (11) after the gap substitution, and no such substitution is available for  $\Delta \lambda^{(B)}$  at general  $\eta$ . (The coherence identity of Proposition 1 does transfer wherever  $u^{(1)}$  is piecewise constant, but no quantity in (12) depends on it.) Second, (37) compares noise tolerance for a *fixed* split; which split each rule targets is the separate question of Remark 8.

## G Supplementary Empirical Figures

Section 5 restricts the main text to one figure per regime, both rounded by  $k$ -means on  $L_{\text{sym}}$ . This appendix collects the sweeps that support claims made there but are not themselves plotted: the balanced baseline, the identity checks behind the generated-regime discussion in Section 5, the full recovery grid, and the cross-operator comparison whose absence from the main text is explained below.

## Balanced baseline

Figure 6 sweeps the balanced case ( $\eta = 1$ ) referenced in the synthesized regime of Section 5, scored by the measured NMI of Corollary 4, consistent with the main-text figures.

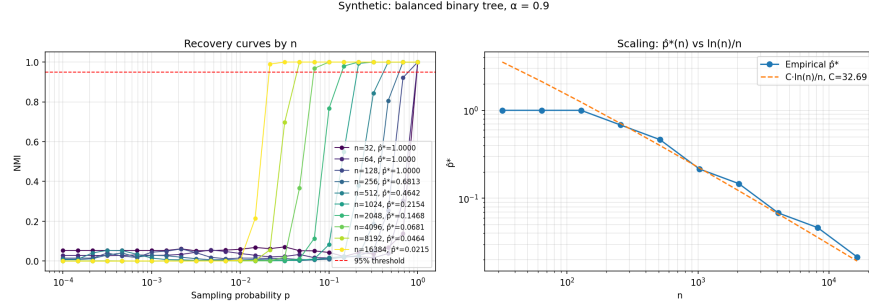


Figure 6: **Synthesized, balanced flat CBM ( $\eta = 1$ ).**  $\hat{p}^*$  (measured  $\text{NMI} \geq 0.95$ ) vs  $n = 32, \dots, 16384$ , following  $\hat{p}^* = C \log n/n$  (dashed). The fitted constant is  $C = [\text{TBD}, \text{C3}]$ , the  $\Theta(\log n/n)$  baseline of (11) at  $\eta = 1$ .

## Identity checks, generated regime

Figure 7 is the scatter behind the generated-regime identity claims of Section 5: measured coherence, Fiedler eigenvalue, and spectral gap against their Proposition 1 and lemma 1 predictions, across the Kingman/JC69 sample.

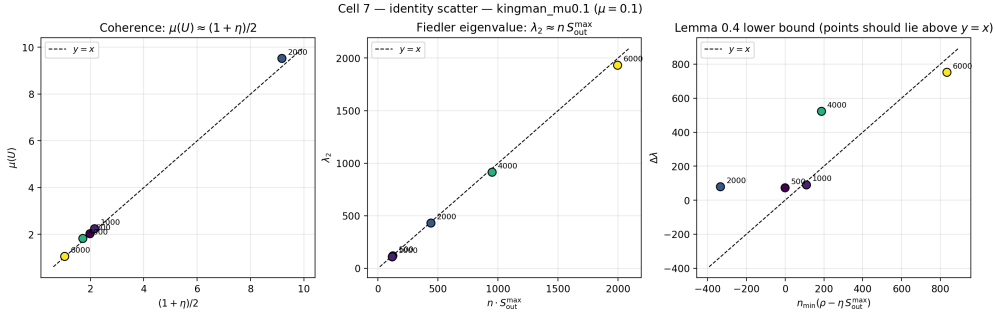


Figure 7: **Generated (Kingman), identity checks.** Measured coherence  $\mu(U)$  against the Proposition 1 prediction  $(1 + \eta)/2$ , and measured gap  $\Delta\lambda$  against the Lemma 1 lower bound  $m_{\min}(\rho - \eta S_{\text{out}}^{\max})$ , on the Kingman/JC69 sample of Figure 3.

## Full recovery grid

Figure 8 is the complete  $\eta \times n$  grid underlying Figure 3: one measured-NMI-vs- $p$  panel per  $(\eta, n)$  pair, rounded by  $k$ -means on  $L_{\text{sym}}$ .

## Rounding-rule sensitivity

Section 5 restricts the main text to  $k$ -means, since a single rounding rule cannot support a claim about rounding rules in general. Figure 9 reruns the same  $\eta$ -pooled sweep of Figure 3 with two further rules applied to the same sampled Fiedler vector: the sign rule that Theorem 1 analyzes, and

Figure 2 — NMI vs  $p$  (rows:  $n$ , cols:  $\eta$ )

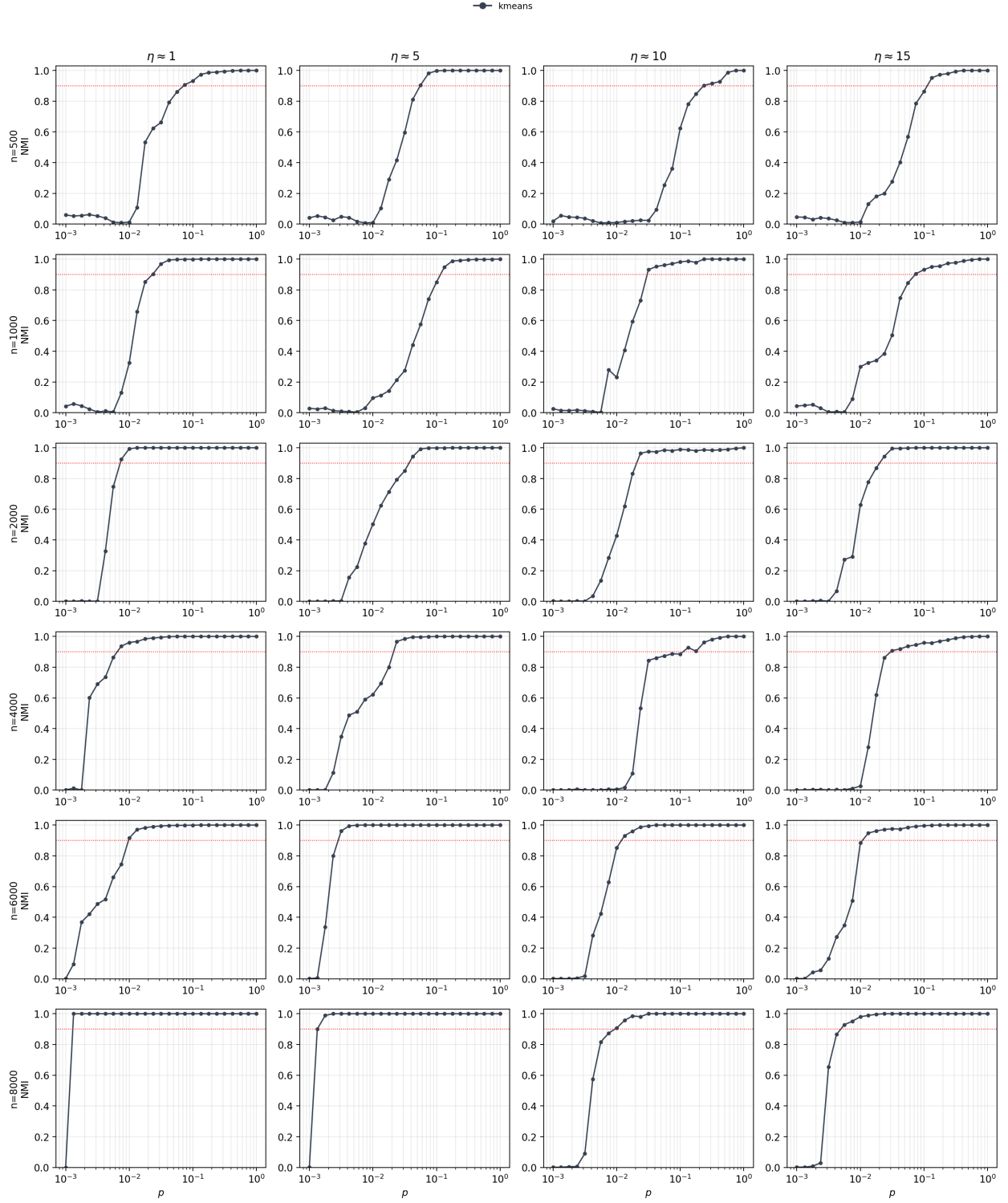


Figure 8: **Generated (Kingman), full recovery grid.** Measured NMI vs  $p$  for every  $(\eta, n)$  pair swept in Figure 3, rounded by  $k$ -means on  $L_{\text{sym}}$ ; the dashed line marks the 0.95 threshold defining  $\hat{p}^*$ .

a  $\sigma_2$ -gap rule. The sign and  $k$ -means operators decay as  $\approx n^{-1}$ , parallel to the  $\log n/n$  reference, while the  $\sigma_2$ -gap rule saturates — so the  $\log n/n$  scaling is a property of the spectral signal, not of the particular rounding rule. This is methodology supporting the choice of  $k$ -means as the main-text operator, not additional evidence for Theorem 1, which concerns the sign rule alone.

Figure 1 —  $p^*$  vs  $n$  by operator, per  $\eta$  (dashed = free power-law fit  $p^* \propto n^b$ )

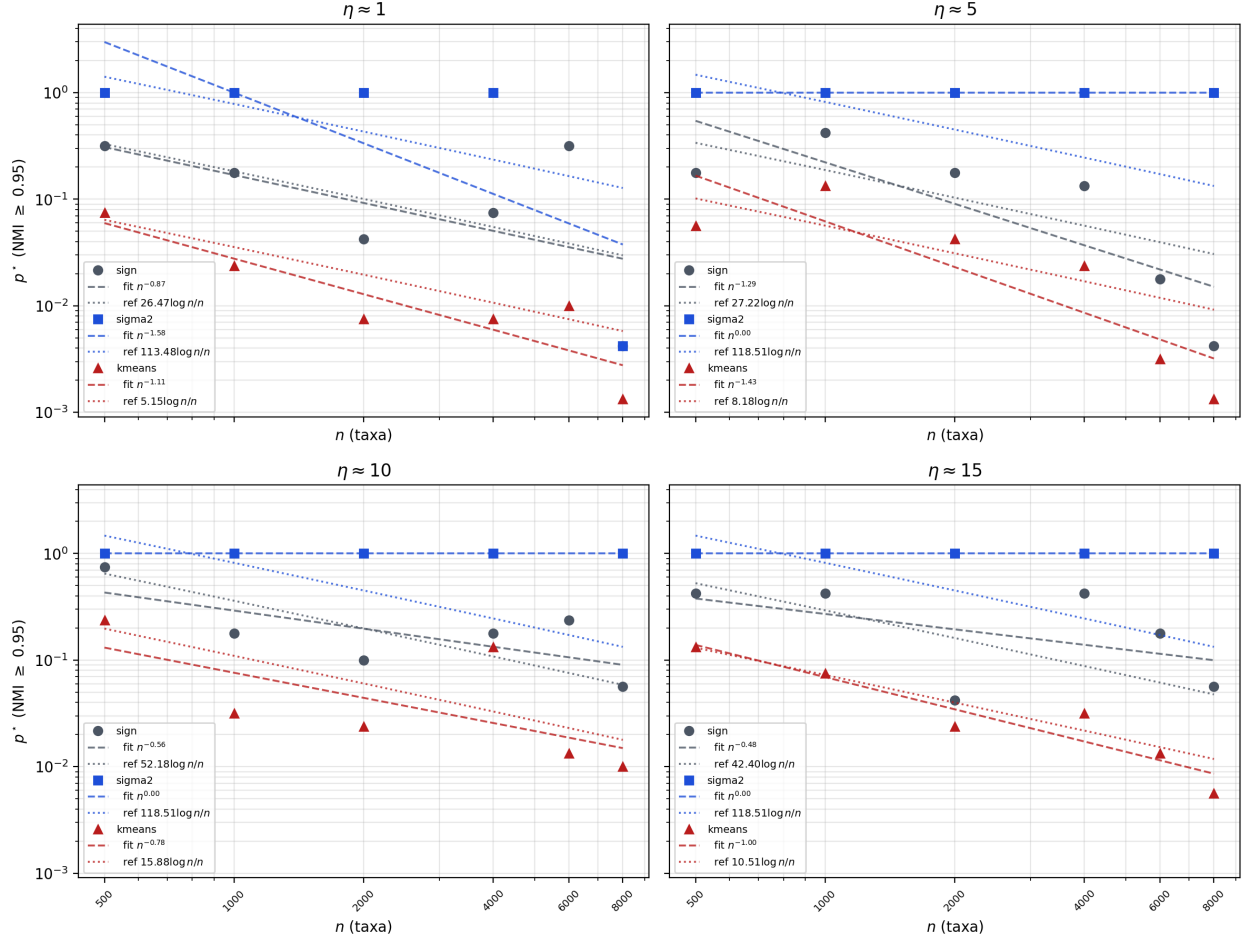


Figure 9: **Generated (Kingman), rounding-rule sensitivity.**  $\hat{p}^*$  (measured NMI  $\geq 0.95$ ) vs  $n$  on the  $\eta$ -pooled samples of Figure 3, one panel per  $\eta \in \{1, 5, 10, 15\}$ , for three rounding rules. Dashed = free power-law fit  $\hat{p}^* \propto n^b$ ; dotted = the  $C \log n/n$  reference. The Fiedler-sign (grey) and  $k$ -means (red) operators follow  $b \approx -1$ ; the  $\sigma_2$ -gap operator (blue) saturates.